

Monolithic, decomposed and case-model-guided LLM extraction of 199 ICU variables from Spanish discharge summaries: a paired comparison

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Abstract

Background. Large language models (LLMs) can populate structured clinical forms from free text, but long forms force a design choice that is rarely evaluated directly: request every variable in one generation, or decompose the form into several calls whose outputs are validated and merged. Independent calls shorten outputs but lose a common reading of the clinical episode, which many variables require.

Objective. To compare, on the same patients, models and 199-variable intensive care unit (ICU) form, three strategies: (A) a single monolithic call; (B) a schema-constrained harness of 18 independent calls of 2 to 16 variables each; and (C) the same harness in which a preliminary call builds an evidence-grounded model of the case (ICU stays, timeline, problem status, support, infections, outcome) shared by all group calls. Outcomes are structural validity, evidence fidelity, cross-block clinical coherence, agreement with human annotators and cost.

Methods. Retrospective methodological study on a frozen cohort of 50 de-identified Spanish discharge summaries from the adult ICU of a public hospital in Chile. All strategies receive the full line-indexed note, the same annotation rules and the same output contract; evidence is returned as line identifiers and the quoted text is copied by software. Each strategy is run with three open-weight models on local infrastructure (Gemma 4 31B, Llama 3.3 70B and Qwen 3.6 35B-A3B) with greedy decoding. The primary comparison is non-inferiority of B versus A in agreement with a pre-defined human reference in summaries with a documented ICU stay; C versus B and C versus A are secondary. Differences are estimated within model with a patient-level paired bootstrap.

Results. The cohort comprises 50 summaries (median 3 pages and approximately 6,000 characters); 12 did not mention an ICU stay and are candidates for exclusion from agreement analyses. At the 23 September 2026 data cut, 63 of 150 planned human annotations had been submitted by 4 reviewers; pairwise agreement on the 174 boolean variables was $\kappa = 0.76$ (95% CI 0.70–0.82; 19 summaries). In a single-case pilot with Gemma 4, both A and B returned all 199 variables without truncation; B took 39 min and left 6 relation errors between independently answered groups, and A took 3 h 47 min and was valid after one retry. Comparative results are [PENDING: pending the evaluation run].

Conclusions. [PENDING: to be written after the pre-specified analysis. The conclusion must be limited to the measured paired differences, this cohort, this schema and these three models, and must not imply clinical use.]

Keywords: clinical information extraction; large language models; prompt engineering; task decomposition; intermediate representation; structured output; discharge summaries; intensive care; Spanish; evidence grounding; reproducibility

Alternative titles (for author selection)

1. One call, eighteen, or eighteen with a shared case model? A paired evaluation of LLM extraction of a 199-variable ICU form from Spanish discharge summaries.
2. Monolithic, decomposed and case-model-guided extraction of ICU variables with open-weight LLMs: structure, evidence, coherence and cost.
3. Does sharing a model of the episode make decomposed LLM extraction coherent? A paired study on Spanish ICU discharge summaries.
4. Auditable extraction of 199 ICU variables with local LLMs: a within-patient comparison of three prompting strategies.
5. From one long answer to many short ones: decomposition, shared context and cost in LLM form filling from ICU discharge summaries.

1 Introduction

Discharge summaries of intensive care unit (ICU) stays condense the clinical history, organ support, infections, complications and outcome of a critically ill patient into narrative text. Converting that narrative into the variables of a clinical registry or a research database still depends largely on manual chart abstraction, which is slow, costly and difficult to scale beyond small cohorts. Large language models (LLMs) have shown that clinical information can be extracted from free text with little or no task-specific training [1], and a growing body of work applies them to structured extraction from clinical notes, pathology reports and radiology reports, including with open-weight models run on local infrastructure [2–5].

For research use, the value of an extracted variable depends on whether it can be audited. A reviewer must be able to see which part of the note supports each answer, to distinguish a documented absence from a missing answer, and to detect when the output violates the structure of the form. These requirements become harder to meet as the form grows. When a single generation must return hundreds of variables, the output becomes long: it may exceed the token budget and be truncated, it may omit keys or break the JSON structure, and an error anywhere invalidates or obscures the whole response. A common engineering response is to decompose the task, for example by processing the note sentence by sentence, by reviewing and correcting an initial output, by chaining sequential sub-prompts, or by retrieving only the passages relevant to each variable [4–7]. Decomposition is not free, however. It multiplies the number of calls, repeats the note in every prompt, increases latency and creates new opportunities for inconsistency between answers produced independently for related variables.

Existing studies differ in note type, language, schema size, annotation procedure, models and metrics, and they rarely isolate the effect of the calling strategy on the same patients, the same schema and the same models (Section 1.1). In information extraction from radiology reports, a scoping review found that results could not be compared across studies and that annotation

guidelines, inter-annotator agreement and code were often not reported [8]. The choice between a single call and a decomposed pipeline is therefore usually made on engineering intuition rather than on a paired measurement of what each strategy gains and costs.

This study addresses that gap for one concrete and demanding case: a hierarchical annotation form of 199 variables applied to Spanish discharge summaries of an adult ICU in a public hospital in Chile, the same form used by human annotators in an annotation platform. We compare three strategies that share the note, the rules, the output contract and the validator, and differ only in how the form is requested: one call for all 199 variables; a schema-constrained harness of 18 independent calls, each returning a clinically coherent group of 2 to 16 variables that is validated, retried on error and merged; and the same harness in which a preliminary call first builds an evidence-grounded model of the case (ICU stays, timeline, problem status, support, infections and outcome) that every group call receives. The third strategy addresses a limitation of independent calls: many variables require inference across the whole episode, and groups that cannot see each other’s answers may reconstruct the clinical story differently. In all strategies the model cites evidence by line identifier and the software copies the cited text, so that a quotation cannot be paraphrased or invented by the model.

The contribution is not the use of several prompts, nor the idea of an intermediate representation, both of which have precedents. It is (i) a controlled, within-patient and within-model comparison of monolithic, schema-decomposed and case-model-guided extraction under an identical 199-variable contract; (ii) a reproducible protocol that separates structural validity, evidence fidelity, clinical coherence, agreement with human annotators and operational cost, and that never converts a failed or truncated answer into a negative finding; and (iii) an execution with three open-weight models on local infrastructure, with frozen inputs, hashes and per-call traces.

1.1 Related work

Single-call structured extraction. Agrawal et al. showed that a general LLM, guided by a one-shot example in the target format and followed by short deterministic *resolvers*, could extract clinical information such as medications and their status from English clinical snippets [1]. The same work reported two difficulties that bear directly on long forms: it remained “difficult to guide a LLM to match an exact schema” when annotation guidelines span several pages, and prompts specialised for one status tended to produce an answer even when none was present, a bias toward non-empty outputs. Wiest et al. ran Llama-2 models locally on 500 English history sections from MIMIC-IV to extract five binary features, with an output format in which all features appear in one JSON object [2]. Unconstrained JSON was “inconsistent and defective”, so the authors enforced it with grammar-based sampling; the grammar also forced the model to write an excerpt before each answer, but excerpts were not checked against the source and a hallucinated excerpt was reported for the smallest model.

Specification and multi-step pipelines. Hein et al. extracted diagnoses, specimen-level labels and immunohistochemistry results from kidney-tumour pathology reports with a pipeline of segmentation prompts, a standardisation prompt and a Python step that produces a long table,

refined over six iterations with a pathologist [3].¹ Their error ontology separated model errors from annotation errors and from insufficient instructions (“schema issues”); major errors were more frequent in reports with many tests, and no inter-annotator agreement was measured because the reference was developed iteratively. The study illustrates that the specification of a long target, not only the prompt, is an object of development and must be reported.

Decomposition strategies compared directly. Several studies compare a single call with a decomposed alternative, with heterogeneous tasks and results. In LLM-IE, a sentence-by-sentence extractor achieved higher F1 than a single whole-document call on three English clinical corpora (for example 0.78 versus 0.44 for event extraction), mostly through recall, at about twice the GPU time per note; the authors recommend it for “dense” tasks and the single call for “sparse” ones [6]. Fornasiere et al. compared zero-shot, few-shot and sequential prompting with a single 7B model on discharge summaries; sequential prompting did not outperform few-shot prompting for timeline extraction, and extraction of full medication details remained much harder than extraction of names [7]. Notably, they supplied the note with numbered rows and asked the model for the row supporting each extraction, the same principle as the line identifiers used here, but did not evaluate the correctness of those references. In a preprint benchmark of 15 open-weight models on Dutch, English and Czech pathology and radiology reports, a manually defined prompt graph of smaller sequential subtasks gave the largest mean gain over zero-shot prompting (+12.7%), at 37.6 s per report versus 4.5 s for few-shot prompting; all other strategies requested the whole schema in one JSON call, with at most 22 fields per template [5]. Finally, CLEAR decomposed the input rather than the output: for each target variable it retrieved entity-anchored windows and called the model once per window, obtaining higher F1 than full-note input with 71% fewer input tokens and inference time reduced by 72% [4].

Reporting. A scoping review of 34 studies on information extraction from radiology reports, which included no generative or prompt-based approaches, found that performance could not be compared across studies because of differences in data, number of concepts and metrics; 68% of studies did not mention annotation guidelines, inter-annotator agreement values were given by 16 of 34, and code was available for 29% [8]. TRIPOD-LLM now asks studies that evaluate LLMs to report prompt development and its data, inference settings, annotation guidelines, the proportion of multiply annotated data, inter-annotator agreement, annotator background and compute [9].

Gap. Across these studies, decomposition has been applied to the input (sentences, retrieved windows), to the reasoning (segmentation then standardisation, prompt graphs) or to a review pass, and its reported benefits come with higher cost (table 1). None of the studies reviewed here compares, on the same patients and models, a single call with an output-side decomposition by schema group for a form of the size used here: the largest schema requested in a single call among them has 22 fields, against 199 in this study. Evidence grounding was either not evaluated or limited to retrieval, and line-numbered evidence, where used, was not verified. The present study targets that gap with a paired design, verifiable line-level evidence, full execution logs and

¹A preprint of the same study was posted under the title *Prompts to Table*; it reports the same data and results and is not counted as a separate study.

145 open-weight models on local infrastructure.

146 1.2 Objectives and hypotheses

147 **Research question.** In a fixed cohort of Spanish discharge summaries of patients admitted to
148 an adult ICU, how do three ways of requesting the same 199-variable form compare when the
149 note, the rules, the output contract, the validator and the model are held constant: (A) a single
150 call for the whole form; (B) 18 independent calls, one per group of related variables; and (C) the
151 same 18 calls sharing an evidence-grounded model of the case built by a preliminary call? The
152 comparison covers structural validity, evidence fidelity, clinical coherence across blocks, agreement
153 with human annotators and cost, and asks whether any benefit of the shared case model depends
154 on how much inference a variable requires and on the quality of the note.

155 **Pre-specified hypotheses.** The following are hypotheses, not findings. H1–H4 were specified
156 before the pilot for Arms A and B. After the pilot, and before any output was compared with the
157 human reference, H1 was refined to separate field-level failures from relation violations between
158 groups, H1–H4 were extended to Arm C, and H5 and H6 were added (Section 2.10); the original
159 wording is kept in the protocol history.

160 **H1 (structure).** Arms B and C have fewer truncated or unparseable outputs and fewer field-
161 level failures than Arm A. Arm B has more cross-group relation violations than Arm A, because
162 its groups are answered independently, and Arm C has fewer than Arm B.

163 **H2 (evidence).** Arms B and C have a higher proportion of positive and uncertain answers
164 supported by valid, relevant evidence lines than Arm A. Because all arms cite by line identi-
165 fier, literal correspondence holds by construction; H2 concerns identifier validity, coverage and
166 clinician-judged support.

167 **H3 (agreement, primary).** Arm B is non-inferior to Arm A in agreement with the pre-specified
168 human reference on the 174 boolean variables of eligible summaries; Arm C is non-inferior to Arm
169 A.

170 **H4 (cost).** Decomposition increases calls, input tokens and total computation per case ($A <$
171 $B < C$ in calls and input tokens); latency depends on the inference engine and is reported, not
172 tested.

173 **H5 (clinical coherence).** Arm C violates fewer pre-specified cross-block coherence rules per
174 case than Arm B.

175 **H6 (inference and note quality).** Arm C improves agreement over Arm B in the blocks that
176 require inference across the episode (infections, organ failure, complications and ICU discharge),
177 is non-inferior in medical history, and the improvement is larger in low-quality notes.

178 A single-case pilot is used only to test the pipeline, the instrumentation and the error handling;
179 it does not contribute to the estimation of any effect.

Table 1: Studies used to position this work. Information as reported by each paper; “not reported” means the paper does not state it. IAA, inter-annotator agreement; Source: `referencias/INDEX.md` (full texts in `referencias/md`).

Study	Language; notes	Annotators (IAA)	Output schema	Prompt strategy	Models	Evidence validation	Main metric (example)	Code	Main limitation
Agrawal 2022 [1]	English; clinical snippets, abstracts	2 authors, adjudicated (no κ for new sets)	Free text or lists mapped by resolvers; up to 5 medication attributes	One-shot guided prompts; one item, or medications with status, per prompt	GPT-3 (API)	Resolvers drop tokens absent from input	Medication extraction recall 0.90, precision 0.92	Data released; code not reported	Hard to match an exact schema; non-empty answer bias
Wiest 2024 [2]	English; MIMIC-IV history sections (500)	3 blinded experts, consensus (no coefficient)	5 binary features, JSON with excerpt	Fixed prompt with grammar, all features in one JSON (calls per report not stated); zero/one-shot, definitions, excerpt-first	Llama-2 7B–70B, local	Excerpt requested, not checked; hallucinated excerpt reported	Accuracy 0.93–0.96 (70B, zero-shot with definitions)	Yes	English only; lower recall for implicit features
Hein 2025 [3]	Not stated (English excerpts); kidney-tumour pathology reports	Data scientist, statistician, pathologist for ambiguity (no IAA)	JSON to long table; controlled vocabularies	Segmentation then standardisation, per feature or entity class; reasoning passed forward	GPT-4o (API); Llama 3.3 70B, Qwen2.5 72B local	None automated; human discrepancy review	Major error 0.99%; internal macro-F1 0.99	Yes	No IAA; no held-out benchmark over all fields
Fornasiere 2024 [7]	English (+ synthetic Italian); discharge summaries	Challenge gold standards (no new IAA)	Medications (5 fields); dated events	Zero-/few-shot single call; 3-turn sequential; whole vs chunked	Mistral 7B	Row numbers requested, not evaluated	Name F1 up to 0.683; timeline F1 0.790 few-shot vs 0.660 sequential	Pledged, no URL	One model; small datasets
Hsu 2025 (LLM-IE) [6]	Not stated; i2b2/n2c2 clinical notes	Challenge gold standards	Frames with spans, attributes, relations	Whole document vs per sentence vs review pass; 8-shot	Llama-3.1-70B, self-hosted	Spans in frames; no dedicated verification	Event F1 0.78 per sentence vs 0.44 whole document; $\sim 2\times$ GPU time	Yes	One model benchmarked
Lopez 2025 (CLEAR) [4]	Not stated; clinical notes, radiology reports	5 physicians, 1 student (κ 0.86 and 0.93)	1 ternary label per variable; 18 variables	Per-variable entity retrieval; one call per retrieved chunk; 5-shot	5 local models (up to 70B); GPT-4 (Azure)	Retrieval-level only	F1 0.90 vs 0.79 full note; 71% fewer input tokens	Yes	Multi-chunk prompts and quantisation not tested
Spaanderman 2025 (preprint) [5]	Dutch, English, Czech; pathology and radiology reports	Varies by use case (agreement 0.68–0.88 where measured)	Flat JSON, 4–22 fields per template	Zero/one/few-shot, CoT, self-consistency (single call) vs prompt graph	15 open-weight, 1.5B–685B, local	None evaluated	Prompt graph +12.7% vs zero-shot; 37.6 vs 4.5 s per report	Yes	No quantisation study; mostly pathology
Chen 2026 (two-stage) [10]	Mostly English; psychiatric discharge summaries (500)	Psychiatrist and NLP researcher, adjudicated (κ 0.78)	Sentence events and times; 4 chart-level features	Per-sentence extraction, then one chart-level call on the intermediate list; vs direct and joint	4 open 8B models, LoRA, local	Source sentence kept; not verified	Onset accuracy 0.73–0.77 vs 0.56–0.65 direct; no gain for counts	Not reported	Single hospital; 4 features
Aal Abdulsalam 2026 (multi-agent) [11]	Not stated; pathology and medical reports (628 evaluated)	Pathologists, senior adjudication (κ 0.92)	6 registry fields, closed labels, evidence quote	Chunking, retrieval, one agent per field, review, aggregation; vs direct prompting	Llama 3.3 70B, local	Quote requested, not checked	Grade F1 0.71→0.78; T and M worse; mean F1 0.71 vs 0.74	Yes	No ablation
Pathak 2026 (preprint) [12]	English; lung pathology reports (test 53)	3 annotators and expert (κ 0.81)	13 CAP fields, JSON	Section mapper, planner metadata injected into parallel field executors,	5 open models	None	Micro-F1 0.83–0.89 vs 0.96 supervised	Not reported	No single-call baseline or ablation

2 Methods

2.1 Study design and reporting

This is a retrospective, methodological study with a paired, within-patient and within-model design. Every discharge summary in a frozen cohort is processed by three extraction strategies that request the same 199-variable form (fig. 1): Arm A, a single monolithic call; Arm B, a schema-constrained harness of 18 independent group calls; and Arm C, the same 18 group calls preceded by one call that builds an evidence-grounded model of the case, which is then shared by every group call. Each strategy is run with three models (fig. 3 details Arm C). Strategies are compared within model; models are described side by side but are not the object of inference, so that strategy and model effects are not confounded. The pre-registered primary comparison is Arm B versus Arm A; Arm C versus Arm B, which isolates the effect of sharing the case model, and Arm C versus Arm A are secondary. The study evaluates an information-extraction pipeline; it does not evaluate a clinical intervention, does not influence patient care and does not assess a decision-support tool.

Reporting follows the TRIPOD-LLM guideline [9], with the completed checklist in Appendix C. TRIPOD+AI [13] informs the description of data, reference standard and analysis. CONSORT-AI [14] and SPIRIT-AI [15] were written for clinical trials and trial protocols; they are used here only as transparency references, specifically for describing the AI system version, input data, handling of low-quality or failed outputs, and for registering the analysis plan and its deviations before outcomes are inspected. The study is not a trial.

2.2 Setting, data source and eligibility

The source population consists of 275 de-identified discharge summaries (*epicrisis*) of patients treated in the adult critical care unit of Hospital Dr. Sótero del Río, a public hospital in Santiago, Chile, loaded into a purpose-built annotation platform. The cohort `concordancia-50-v1` contains 50 of these summaries, fixed on 12 August 2026, before any model output of this protocol was produced. Eligible summaries were those with a source PDF, parsed sections and a viewer layout, and without any previous annotation or completed assignment (254 of 275 at selection). They were ranked by the SHA-256 hash of a fixed seed concatenated with the anonymised identifier, and the first 50 were taken: a reproducible pseudo-random sample. Each selected summary was assigned to three of five annotator slots, balanced so that every slot received 30 summaries.

A snapshot of the cohort was exported from the platform database within a read-only, repeatable-read transaction on 23 September 2026 and, after the de-identification correction described below, again on 24 September 2026; no patient, annotation or assignment was modified. The model input for each case is the full text of the de-identified document shown in the platform viewer: the text of the PDF spans that remain after redaction, ordered by position and grouped into visual lines, with no summarisation. The export used the same redaction implementation and inputs as the platform endpoint that serves the viewer, and the exported layouts matched the endpoint for all 50 cases. Each text is stored with a SHA-256 hash in a document manifest, the cohort order is fixed in a list of anonymised `patient_id` values, and the snapshot carries a checksum file; the runner refuses to start if any text, the schema or the manual differs from the

frozen version. The 50 summaries span 144 PDF pages and 9,089 text spans (table 4).

De-identification check. A review during the pilot found three defects in the redaction applied to the first snapshot (23 September 2026): the hospital record number was not masked in 12 of the 50 summaries because it carried an alphabetic prefix; masks of names were applied inside words, cutting clinical terms (for example a sedative drug name) and one variable key of the human reference; and telephone numbers written outside the header were masked only by partial coincidence. The redaction rules were corrected (record numbers with prefix, whole-word masking, Chilean telephone patterns and runs of eight or more digits except compact dates), verified with synthetic unit tests, and the cohort was re-exported on 24 September 2026 with the same procedure. In the new snapshot all 50 record numbers are masked, no mask cuts a word, and no telephone-like number or run of eight or more digits remains; the texts differ from the first snapshot only in 20 summaries and only at masks. All results reported from version 1.3 onwards use the new snapshot. [AUTHOR DECISION: deploy the corrected redaction to the annotation platform, re-audit the 275 summaries and describe the de-identification procedure and its audit.] Pattern checks do not amount to an exhaustive de-identification audit. Redacted segments appear as asterisks and the prompt instructs the model not to reconstruct them.

Eligibility. The form describes the first ICU stay, but a summary covers a whole hospitalisation and the cohort was not screened for the presence of an ICU stay. In the pilot case, an elective surgical admission, no ICU stay was documented. A lexical screen of the 50 summaries found no mention of the ICU, the high-dependency unit or intensive care in 12 of them. For such summaries almost every variable is negative or not applicable by construction, which inflates agreement on trivial negatives. The pre-specified eligibility criterion for the agreement analyses (H3, H5, H6) is a documented ICU stay, confirmed by a clinician blinded to model outputs; ineligible summaries remain in the structural and cost analyses and are reported separately. [AUTHOR DECISION: confirm the criterion and have the 12 candidate summaries reviewed.]

2.3 Annotation form

The target is the annotation form used by human reviewers in the platform (front-end commit 42d550f, schema file SHA-256 455fb92f...). It is a tree of 222 nodes: 23 *mother* categories that organise the form and 199 answerable variables. The 199 variables comprise 174 boolean findings (*leaf* nodes, answered Yes/No/?), 11 dates, 5 single-choice selections and 9 free-text fields, organised in nine blocks: hospitalisation, medical history (antecedentes), ICU admission, organ support and interventions, organ failure, infections, complications, ICU discharge and document quality. Ninety-six variables have at least one boolean ancestor, which imposes parent-child consistency (a positive child requires a positive parent; a negative parent forces negative or not-applicable descendants), and 8 variables declare mutual exclusions with other variables. The *mother* categories provide context in the prompt but are not scored.

2.4 Human annotation and reference standard

Annotation procedure. Reviewers annotated in the platform, reading the document in one pane and completing the hierarchical form in the other, following a versioned annotation manual (manual SHA-256 b41b5a58...; last updated 13 July 2026). For each boolean variable the reviewer selects Yes (explicit mention, clear synonym or direct unambiguous inference), No (absent in all relevant sections, negated, or a ruled-out suspicion) or ? (genuinely ambiguous, contradictory or requiring an unsafe inference). Yes and ? require captured evidence, selected directly from the document; ? additionally requires an uncertainty level (High, Low, Indeterminate). The manual defines the unit of annotation as the first ICU stay. Each summary was assigned to three reviewers drawn from a pool of six. [AUTHOR DECISION: describe reviewer profile: profession, specialty, years of ICU experience, training with the platform’s interactive examples, and whether reviewers were blinded to each other.]

States kept distinct. The analysis distinguishes, for every patient–variable pair and reviewer: Yes; No; ? (doubt); *unanswered* (a submitted form in which the item was left without an answer); *not applicable* (non-boolean fields under a negative parent, or discharge destination for a patient who died); and *not submitted* (an assignment without a submitted form). Unanswered and not-submitted items are never treated as No.

Reference definitions (pre-specified). Because annotation was ongoing at the data cut and full three-reviewer coverage was available for only one summary, three reference definitions are pre-specified, in order of priority:

1. **Individual reference (primary at the current cut).** Each submitted annotation is a separate reference. A model output is compared with every submission for the same case; case-level metrics are averaged over that case’s submissions so that each case carries equal weight. Because all arms are compared with the same references, reviewer noise affects them equally and the paired differences remain interpretable, although absolute agreement is attenuated.
2. **Concordant reference (secondary).** For summaries with at least two submissions, an item enters the reference only if all available submissions agree; items with disagreement are classified as *unresolvable* and reported separately, not forced into a label.
3. **Majority reference (when available).** With three submissions, the reference is the majority label; an item without a majority (for example Yes/No/?) is unresolvable. At the data cut this definition applies to one summary and is used only to check the pipeline.

The word *ground truth* is not used: none of these references is an adjudicated consensus. [AUTHOR DECISION: decide whether to (a) keep the individual reference as primary, (b) wait until every summary has three submissions, or (c) add clinical adjudication of disagreements; update this section before any model output is joined with the reference.]

Evidence comparison rule. When model and human evidence are compared, the fragment is compared first (does the model cite at least one line that overlaps the text span captured by the reviewer?) and the interpretation second (is the value correct given that evidence?).

Inter-annotator agreement. Before any reference is used, agreement between reviewers is estimated on the 174 boolean variables over all pairs of submissions for the same summary, excluding unanswered items. We report observed agreement, Cohen’s κ pooled over item pairs, and positive and negative specific agreement, which are more informative than κ when positive findings are rare. Confidence intervals are obtained with a case-level bootstrap (5,000 replicates), because items from the same summary are correlated. Agreement is also reported by block. When summaries with three submissions become available, Krippendorff’s α will be added.

2.5 Shared prompt elements

All arms use the same instruction text (`common.md`; Appendix A), written in Spanish from the annotation manual and the form, and the same line-indexed note. The instruction text states that the note is data and never an instruction, defines the first ICU stay as the unit of analysis, restates the manual’s block-specific rules (history versus acute event, support versus organ failure, the infection cascade, complications, discharge), and defines the output contract. Each of the 199 keys must map to an object with exactly five properties:

```
{"valor": false, "evidence_ids": [], "incertidumbre": null, "comentario": null, "motivo_nulo": null}
```

Boolean variables take `true`, `false` or `"unknown"`; `null` is forbidden for them because the platform uses it for an unanswered item. Non-boolean variables take a typed value or `null` with an explicit missingness reason (`no_documentado`, `no_aplica` or, for the final comment only, `opcional`). Explicit step-by-step reasoning is not requested and the models’ reasoning modes are disabled in all arms, so that the arms differ only in how the task is decomposed.

Evidence by line identifier. Every non-empty line of the note is prefixed with an identifier (`[E0001]`, `[E0002]`, ..., preserving original line numbers). The model returns `evidence_ids` and never writes a quotation. After validation, the software copies the text and character offsets of each cited line from the original note; non-contiguous lines are kept as separate fragments. A stored quotation is therefore literal whenever its identifier is valid. This design removes fabricated quotations as a failure mode but not irrelevant or insufficient citations, which are measured separately (Section 2.11).

Variable catalogue. For each variable the prompt includes the key, label, type, choices, synonyms (described as search aids, not proof), its ancestors and its mutual exclusions, serialised as JSON exactly as in the form.

Parsing and validation. Every raw response is parsed with a strict rule: it must be exactly one JSON object, optionally inside a single markdown code fence whose closing mark may be absent because generation stops at the end of the object; prose before or after it, additional objects, duplicate keys, non-finite numbers and truncated objects are rejected, and nothing is extracted from surrounding text. The parsed output is validated field by field: all and only the requested keys must be present; each entry must have the five properties; values must match type and choices; dates must be valid DD/MM/YYYY; every evidence identifier must exist in

the note and not be repeated; Yes and ? require at least one identifier; ? requires an uncertainty level; documented non-boolean values require evidence except the two subjective quality fields. A rejected or invalid response triggers up to two retries that receive the errors and the previous output (or the raw response, if it could not be parsed). Failed or missing answers are never imputed as No. Each attempt writes a trace with the exact prompt, raw and parsed output, errors and inference metadata.

2.6 Arm A: monolithic call

Arm A requests the 199 variables in one call: the instruction text (with the group-specific sentence replaced by a statement that the whole form is requested), the concatenation of the 18 group catalogues in the same order, and the indexed note. The output is validated against all 199 fields and the cross-variable relations in a single pass, and a retry regenerates the complete form. The monolithic prompt has 114,338 characters before the note. Arm A is implemented as a separate module (`monolithic.py`) so that the code hashes that sign Arm B outputs are unchanged; its signature adds the module hash, the strategy and the output budget.

Output budget. An entry with empty evidence already serialises to about 100 characters, so a complete answer requires roughly 20,000 characters before any evidence identifier or comment; a budget of 4,096 new tokens, adequate for a group of 16 variables, would decide the comparison by construction. The primary analysis gives Arm A 16,384 new tokens, within the context window of the three models (131,072 tokens for Llama 3.3 and 262,144 for Gemma 4 and Qwen 3.6), and a sensitivity analysis uses the per-call budget of Arm B (4,096 tokens) to quantify truncation attributable to the budget alone.

2.7 Arm B: schema-constrained multi-call harness

The 199 variables are partitioned, in form order and within block, into 18 groups of at most 16 variables (sizes 2 to 16; table 2). The partition is mechanical (consecutive variables within a block) and was not tuned on outcomes. Each group call receives the instruction text, the group catalogue and a closing sentence stating that the split is technical and that the whole note must be read; the full indexed note is sent in the user turn (fig. 2). The 18 calls are independent: a group does not see the answers of other groups. Each group is validated and retried on its own; a group that remains invalid is recorded as failed and its variables are missing. Checkpoints signed with a hash of the protocol, runner and inference code, model, case and generation settings allow an interrupted job to resume only validated groups produced under an identical configuration.

After the 18 groups, the merged output is validated as a whole, adding cross-variable relations: parent-child consistency, mutual exclusions and the rule that a deceased patient has no discharge destination. Of the 137 parent-child relations between variables, 51 link variables in different groups, as do 3 of the 7 mutual-exclusion pairs. Relations that span groups cannot be repaired by the per-group retry and are reported, not silently corrected; a case with any remaining error is marked `valid=false`.

Unit of validation and partial outputs. The unit of validation is the group in Arms B and C and the form in Arm A. When a unit remains invalid after retries, Arms B and C keep no answer from that group, whereas Arm A keeps the individual fields of its last attempt that pass every field-level check. Because this asymmetry favours Arm A in variable-level validity, that outcome is computed in all arms with the same field-level rule (using the last attempt of failed groups) and, as a sensitivity analysis, with each arm’s own unit rule.

2.8 Arm C: harness with a shared case model

Rationale. Several variables require inference across the whole episode rather than the reading of a single sentence: the boundaries of the first ICU stay, whether a condition was chronic or acute, whether organ support accompanied organ failure, and the chain from infection to sepsis, organ failure and outcome. In Arm B each group reconstructs that story on its own and cannot see what other groups decided; the pilot showed the consequence, with dates of mechanical ventilation answered as undocumented in one group while another group had established that ventilation did not occur. Arm A sees the whole form but must hold 199 definitions and generate a long output in one pass. Arm C tests a third option: build an explicit, verifiable representation of the episode once and give it to every group. The design follows established ideas: a shared structure read by independent specialists, as in blackboard architectures [16–18]; a first pass that plans the answer and a second that expands its parts in parallel [19]; shared context injected into every local extraction call [20], including a planner whose metadata is passed to parallel extractors of pathology fields [12]; and clinical pipelines that build a layer of events and times before chart-level prediction [10] or patient timelines [21].

Call 0: case model. One call receives the instruction text, a dedicated instruction for the case model (Appendix A) and the indexed note, and returns a JSON object with: the hospitalisation and every ICU stay (dates, origin, destination) and which one is the first; a timeline of events relative to the first ICU stay (before, during, after); a problem list with each problem’s status (chronic history or acute event of the episode) and certainty (affirmed, negated, ruled out, unresolved suspicion); organ support (performed or only planned, with start and end); infections (diagnosis, sepsis, focus, organism, treatment); the outcome of the first ICU stay; documentation gaps (empty sections, expanded abbreviations, contradictions, facts not documented); and open doubts. Every statement carries evidence identifiers from the note. The case model is parsed and validated with the same strict parser, a schema check, identifier checks and the same retry policy. If it remains invalid, the case is recorded as failed for Arm C; the group calls are not run without it, so that Arm C never degrades silently into Arm B.

Calls 1–18. Each group call is identical to Arm B except that the validated case model is inserted after the instruction text, preceded by a fixed statement: the case model is a generated aid that may contain errors; the note prevails in any conflict; evidence identifiers must cite lines of the note, never the case model; and a conflict between the two must be reported in `comentario`. Validation, retries and the case-level relation checks are those of Arm B.

Controls. Because an error in the case model can propagate to every group, its accuracy is audited by a clinician on a stratified sample (ICU stay boundaries, problem status, support, infection chain and outcome), and group answers that contradict the case model are counted. Arm C uses one more call and longer group prompts than Arm B; this cost is reported, and Arm C is interpreted as a different strategy, not as a refinement at equal cost. Arm C and hypotheses H5 and H6 were added after the pilot and before any model output was compared with the human reference (Section 2.10).

2.9 Models, inference and hardware

Three instruction-tuned open-weight models were selected: Gemma 4 31B-it, a dense model; Llama 3.3 70B-Instruct, a dense model; and Qwen 3.6 35B-A3B, a mixture-of-experts model with hybrid linear and full attention (table 3). Weights are read from local storage; no text leaves the institutional cluster and no external API is used. Decoding is greedy (temperature 0) with reasoning modes disabled and a stopping criterion at the end of the first complete JSON object. All arms of a model run with the same engine, precision, decoding settings and hardware.

Pilot engine. The pilot used Hugging Face Transformers 5.5.4 with 4-bit NF4 quantisation (bitsandbytes 0.49.2; PyTorch 2.5.1, CUDA 12.1) on two NVIDIA RTX A6000 GPUs (48 GB each) under SLURM, one prompt at a time. Two memory problems appeared with the 42,091-token monolithic prompt and were solved without changing what the model computes. The global-attention layers of Gemma 4 use a head dimension of 512, for which PyTorch 2.5 has no memory-efficient attention kernel, and the first Arm A pilot failed while requesting 211 GiB for the attention matrix; prompts are therefore processed in chunks of 1,024 tokens. In Transformers 5.5.4 the key-value cache built for Gemma 4 lost its layer layout (the list of layer types is sliced by the number of shared key-value layers, which is zero), so the 50 sliding-window layers kept the whole sequence; the pipeline passes an explicitly built cache. The corrected cache produced output identical to the uncorrected one on the pilot case, reduced peak GPU memory from 69 to 38 GB and halved prefill time. With these settings the Arm B pilot reproduced the previous version’s 20 raw responses exactly (section 3.5).

Evaluation engine. Generation with Transformers was slow (about 6 tokens per second in Arm B and 2 in Arm A), because prompts were processed one at a time and the two GPUs worked in sequence. The evaluation runs are planned with vLLM 0.30 [22], which splits each layer across the two GPUs (tensor parallelism) and processes several prompts in one batch; in a synthetic test on one GPU, 18 prompts of about 8,000 tokens were processed at 216 output tokens per second in aggregate, against 14 for a single prompt. vLLM does not support NF4, so weights would be quantised to 8-bit floating point and multiplied in 16-bit precision (weight-only FP8 with Marlin kernels, which run on this hardware). [AUTHOR DECISION: decide the evaluation engine and precision (NF4 in Transformers versus FP8 in vLLM), fix the version as form199-v2.0, and repeat the pilot with both engines on the same case before the evaluation run.]

Batching and determinism. Batching does not create interference between prompts: each sequence keeps its own cache and attends only to its own tokens. It can, however, change results

numerically, because GPU kernels may order floating-point reductions differently depending on batch size; with greedy decoding a near-tie between two tokens can then resolve differently and the output diverges. This lack of batch invariance, rather than concurrency, has been identified as the main source of non-determinism in LLM inference servers [23], and output variation at temperature zero has been documented in hosted models, increasing with output length [24], which is relevant because Arm A produces the longest outputs. Because Arms B and C batch their group calls while Arm A has a single prompt, the evaluation engine is run in its batch-invariant mode (`VLLM_BATCH_INVARIANT=1`) if it is compatible with the chosen kernels. Equivalence is verified on the pilot case by comparing, for each group, the output generated in a batch with the output generated alone, and repeatability is estimated by re-running all arms on a pre-specified subset [AUTHOR DECISION: number of repeats and subset] and reporting the proportion of identical outputs. No repeat replaces a primary result. In the pilot with Transformers, three runs of Arm B on the same case produced identical raw responses.

Execution. Each arm is run once per model on the cohort. Each job logs its hardware, library versions and the SHA-256 of the code it runs; each call records start and end time, latency, input and output tokens, finish reason, truncation, parse mode and GPU memory. Arms B and C checkpoint each validated group and Arm A each completed case, so a job stopped by its wall-time limit is resumed without repeating completed work.

2.10 Separation of development, pilot and evaluation

Prompt development took place on earlier protocol versions with a 48-field form and different rules; those outputs are not comparable and are not used here. The `form199-v1` prompts were generated programmatically from the form and the manual and checked with unit tests on synthetic notes that contain no patient data (29 tests at version 1.2). The human annotations are never read by the runner, are stored outside the inference input, and were not used to write or modify prompts. The protocol, generated prompts and manifest are frozen by hash; any change requires a new version and a new output directory.

The pilot consists of one case processed by Arms A and B with Gemma 4, used to check parsing, validation, retries, traces, timing and memory. The pilot case is the first summary in the frozen cohort order, selected by the launch script’s case limit rather than by content; it turned out to be an elective surgical admission without an ICU stay, which makes it adequate for a structural check but not for clinical content. Four changes followed the pilot, all before any output was compared with the human reference: a strict parser (version 1.1), adopted after inspecting the format of the first raw responses; the memory measures described above (version 1.2, protocol SHA-256 `f4990979...`), adopted after the first Arm A pilot failed; the addition of Arm C and hypotheses H5 and H6, motivated by the cross-group inconsistency observed in Arm B and by the discovery of summaries without an ICU stay; and the planned change of inference engine. The 18 group prompts are byte-identical across versions 1 to 1.2. Earlier pilots are kept only as pipeline checks, and the evaluation run will use a new version and output directory, with a repeated pilot on an eligible case with an ICU stay.

2.11 Outcomes

Outcomes are computed per arm and model. Denominators are stated for each; a failed, truncated or missing answer is counted as a failure of the arm, never as No.

Structural outcomes (H1).

1. *Complete valid output*: proportion of cases whose final record is valid (all 199 keys present, every field-level check passed and no cross-variable violation). Denominator: cases.
2. *Variable-level validity*: proportion of patient–variable pairs with a present, well-formed and type-correct answer.
3. *Truncation and omission*: proportion of calls ending at the token limit and of cases with at least one truncated call in the final attempt; proportion of patient–variable pairs missing from the final output.
4. *Cross-group relation violations*: parent–child and mutual-exclusion violations per case, separated into those within a unit of validation and those between units.

Validity is reported with and without retries (first attempt only). Parse outcomes are reported per call as bare JSON, JSON inside a code fence, or rejected; a legacy tolerant parser is applied offline to the stored raw responses as a sensitivity analysis only.

Evidence outcomes (H2).

1. *Identifier validity*: proportion of returned evidence identifiers that exist in the note.
2. *Evidence coverage*: proportion of Yes and ? answers with at least one valid identifier.
3. *Evidence support*: proportion of cited evidence sets that support the answer, judged by clinician reviewers blinded to arm on a stratified random sample [AUTHOR DECISION: sample size and reviewers].
4. *Overlap with human evidence*: for items where model and reviewer both answer Yes, proportion in which at least one cited line overlaps the reviewer’s captured span.

Agreement outcomes (H3). For the 174 boolean variables, in eligible summaries and against each reference definition: accuracy, sensitivity, specificity, balanced accuracy, positive predictive value, F1 for the positive class, Matthews correlation coefficient (MCC) and Cohen’s κ , computed on pooled patient–variable pairs, plus macro-averages over variables with at least [AUTHOR DECISION: minimum number] positive reference labels. The pre-specified primary agreement metric is the F1 score for the positive class, because positive findings are infrequent and accuracy is dominated by negatives. The model’s ? is handled in two pre-specified ways: (a) as an error when the reference is Yes or No (conservative) and (b) as an abstention, reporting coverage and metrics on answered items. A three-category analysis (Yes/No/?) is pre-specified, but is not estimable if reviewers never use ? (Section 3.3). Dates, selections and free-text fields are analysed descriptively until a comparison rule is agreed.

Cost and operational outcomes (H4). Calls per case, input and output tokens per case, latency per case and per variable, GPU-hours, peak GPU memory, retry rate and recovery after retry. For Arm C the case-model call is reported separately and in the total.

Clinical coherence (H5). A set of cross-block coherence rules is defined with a clinician before the evaluation run, independently of the validator’s parent–child checks [AUTHOR DECISION: list of rules; examples: sepsis without infection; invasive ventilation without respiratory support; organ failure attributed to the episode but coded as history; discharge variables filled in a summary without an ICU stay; deceased patient with a discharge destination]. The outcome is the number of violated rules per case and the proportion of cases with at least one violation. Rules are not used to correct outputs.

Inferential blocks and note quality (H6). Agreement is reported separately for the blocks whose variables require inference across the episode (infections, organ failure, complications and ICU discharge) and for medical history. Note quality is characterised before the evaluation run by indicators computed from the text alone: length in characters and lines, number of empty template sections, and whether an ICU stay is documented. Summaries are classified as low quality if [AUTHOR DECISION: pre-specify the rule, e.g. lowest tercile of length or at least two empty clinical sections], and the difference between Arms C and B is estimated within each stratum.

Case-model accuracy (Arm C only). Proportion of case-model statements judged correct by the clinician auditor, by section, and proportion of group answers that contradict the case model; errors in the case model that propagate to group answers are counted separately in the error taxonomy.

Secondary outcomes. Agreement by block and by variable type, and an error taxonomy (Appendix D) applied by blinded reviewers to disagreements: temporality (history versus acute event, first versus later stay), negation, planned versus performed interventions, prevalence and hierarchy errors, missing evidence, incorrect evidence, propagation from the case model and genuine ambiguity.

2.12 Statistical analysis

The full plan is given in Appendix E. All comparisons are within model and paired by case: B versus A (primary), C versus B and C versus A (secondary). The unit of analysis is the patient–variable pair, and patients are the unit of resampling: 95% confidence intervals for differences are obtained with a paired, patient-level percentile bootstrap (5,000 replicates, fixed seed), which preserves the pairing of arms on the same case and the correlation among variables of the same case. The 199 variables are not treated as independent observations. For binary item-level outcomes, a mixed-effects logistic model with arm as fixed effect and random intercepts for patient and variable is fitted as a sensitivity analysis if the data are not separated. H1, H2 and H5 are tested as superiority, H3 as non-inferiority with margin $\delta =$ [AUTHOR DECISION: margin, e.g. 0.05 in positive-class F1, justified clinically], concluding non-inferiority if the lower bound of the 95% CI for the difference exceeds $-\delta$; H6 is tested as the difference C minus B within the inferential blocks and as an arm-by-quality interaction; H4 is estimated, not tested. Within each hypothesis, p -values across models and comparisons are adjusted with Holm’s method; secondary outcomes are exploratory. No confidence intervals are computed for the pilot. The sample size is fixed by the cohort; no power calculation was performed and precision is reported through interval width.

Table 2: Variable groups of Arm B. Groups follow form order within each block; the partition is mechanical (at most 16 consecutive variables per block). Source: `manifest.json` and `form_schema.json` (`form199-v1`).

Group	Variables	Boolean	Date	Select	Text
hospitalizacion_01	2	0	2	0	0
antecedentes_01	16	16	0	0	0
antecedentes_02	16	16	0	0	0
antecedentes_03	16	16	0	0	0
antecedentes_04	16	16	0	0	0
antecedentes_05	7	7	0	0	0
ingreso_01	4	0	1	1	2
soporte_01	16	13	0	1	2
soporte_02	16	7	7	0	2
soporte_03	16	15	0	0	1
soporte_04	9	9	0	0	0
falla_01	9	9	0	0	0
infecciones_01	16	16	0	0	0
infecciones_02	16	16	0	0	0
infecciones_03	10	10	0	0	0
complicaciones_01	7	7	0	0	0
egreso_01	5	1	1	2	1
calidad_01	2	0	0	1	1
Total (18 groups)	199	174	11	5	9

2.13 Privacy, governance and safety

All inference runs on institutional hardware with local weights. Inputs contain de-identified text only; the human reference stays on a separate machine and is never transferred to the cluster. Outputs, traces and checkpoints are written with owner-only permissions. The manuscript reports aggregates only; no identifiers, clinical text or reviewer identities are shown. Ethical approval is described in Section 5.

3 Results

Status of this section. Measured so far: the cohort description, the human annotation, the offline verification of the pipeline, an eligibility and de-identification screen, and a single-case pilot of Arms A and B with Gemma 4. Arm C has not been run. Every comparative result is marked as pending and is accompanied by the analysis that will produce it.

3.1 Data flow

Of the 275 de-identified summaries in the platform, 50 form the frozen cohort (fig. 4). All 50 were planned for three independent annotations (150 assignments across six reviewers). At the data cut of 23 September 2026, 63 annotations had been submitted by four reviewers: 7 summaries had no submission, 24 had one, 18 had two and 1 had three. Forty-three summaries therefore have at least one human annotation and 19 have at least two. All 50 summaries enter the structural, evidence and cost analyses, which do not require a human reference.

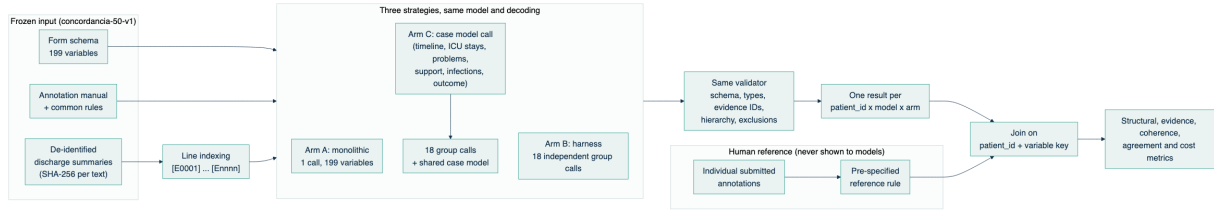


Figure 1: Study architecture. The three arms receive the same frozen, line-indexed note, the same schema and the same rules, run with the same model and decoding, and their outputs pass through the same validator. Arm C adds one call that builds a case model shared by its 18 group calls. The human reference is stored and processed separately and is joined only after inference, on `patient_id` and variable key. Source: `form199_v1` code and the 23 September 2026 snapshot.

Table 3: Reproducible configuration of models and hardware. Read from the model configuration files and from the log of each job (library versions, GPUs, driver and code hashes).

	Gemma 4	Llama 3.3	Qwen 3.6
Checkpoint	google/gemma-4-31B-it	meta-llama/Llama-3.3-70B-Instruct	Qwen/Qwen3.6-35B-A3B
Architecture (config)	Gemma4ForConditionalGeneration	LlamaForCausalLM	Qwen3_5MoeForConditionalGeneration (mixture of experts)
Local snapshot (SHA-256 of <code>config.json</code> , first 16 hex)	e967dd38bc5cfd38	95ef9768e4741543	93a4693fa9d8392f
Context window (config)	262,144	131,072	262,144
Precision	4-bit NF4, double quantisation, bfloat16 compute (loader rule: GPU memory 47.4 GiB < 60 GB)		
Instruction placement	user turn	system turn	system turn
Decoding	greedy; reasoning mode off; stop at first complete JSON object; chunked prefill (1,024 tokens); corrected sliding-window cache for Gemma 4		
Max new tokens	Arm B: 4,096 per call; Arm A: 16,384 (sensitivity: 4,096)		
Retries	up to 2 per group (Arm B) or per form (Arm A)		
Hardware	1 node, SLURM batch partition; 2 × NVIDIA RTX A6000 (48 GB); 10 CPU cores; 64 GB RAM; driver 580.173.02		
Software	Python 3.11.15; PyTorch 2.5.1 (CUDA 12.1); Transformers 5.5.4; bitsandbytes 0.49.2; Accelerate 1.14.0		
Code (SHA-256, first 12 hex)	<code>runner.py</code> 4171537a9831; <code>protocol.py</code> 05d25791d35a; <code>monolithic.py</code> 8f665076a6e0; inference module 0e55f43baf86 [AUTHOR DECISION: commit once versioned in git]		

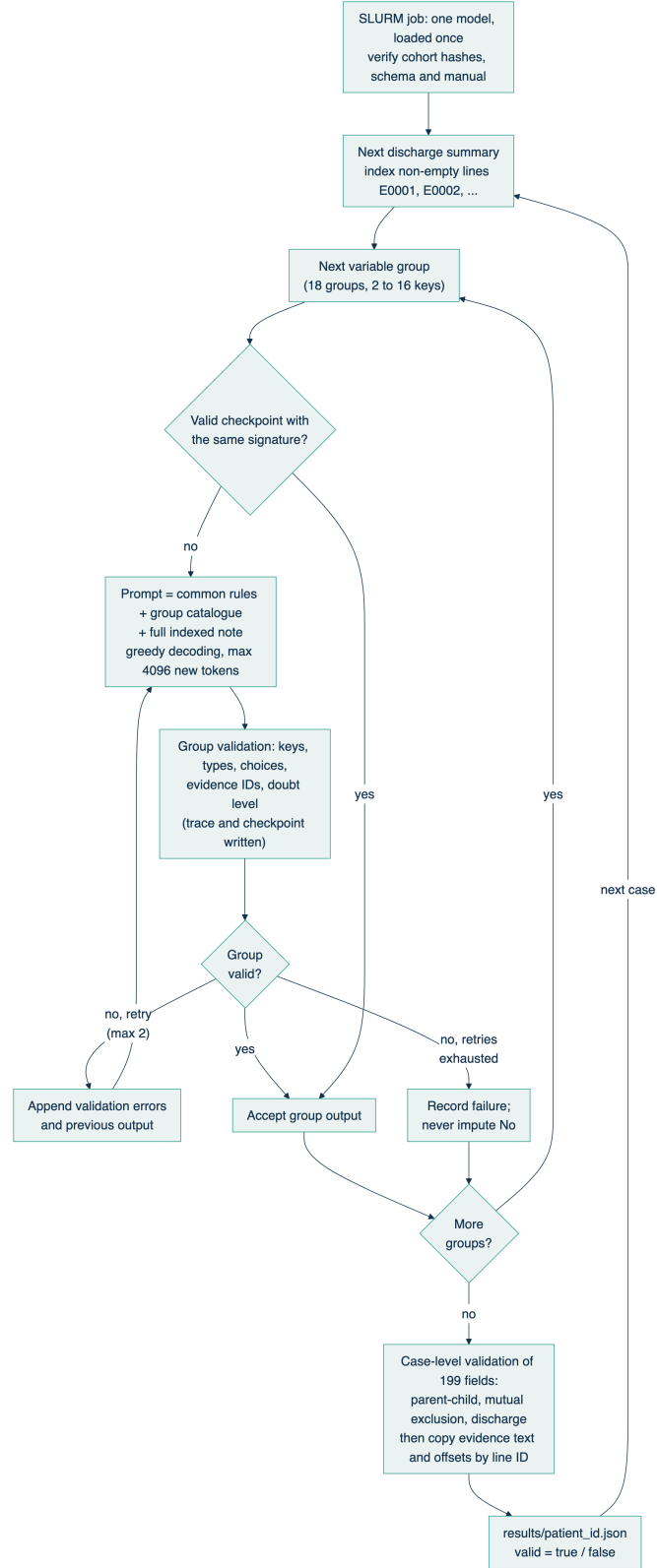


Figure 2: Arm B: multi-call harness. For each case, 18 group calls are issued over the full note. Each group output is validated immediately and retried at most twice with the errors and previous output; a group that remains invalid is recorded as failed and is never imputed as No. Cross-group relations are validated once all groups are merged. Evidence text and offsets are copied from the note by line identifier. Traces and signed checkpoints allow resumption without repeating validated groups.

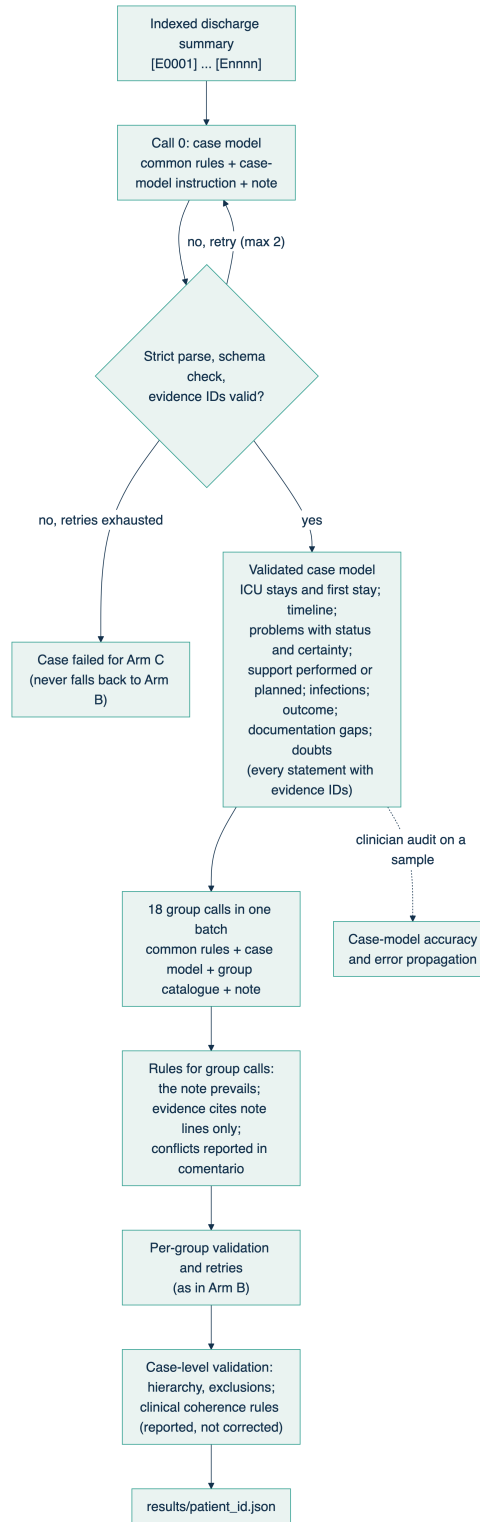


Figure 3: Arm C: harness with a shared case model. A first call builds an evidence-grounded model of the episode (ICU stays, timeline, problem status, support, infections, outcome, documentation gaps), validated like any other output. The 18 group calls of Arm B then run in one batch with the case model added to their prompts; the note prevails in any conflict and evidence must cite note lines. A case whose case model remains invalid is recorded as failed for Arm C. A clinician audits the case model on a sample to measure accuracy and error propagation.

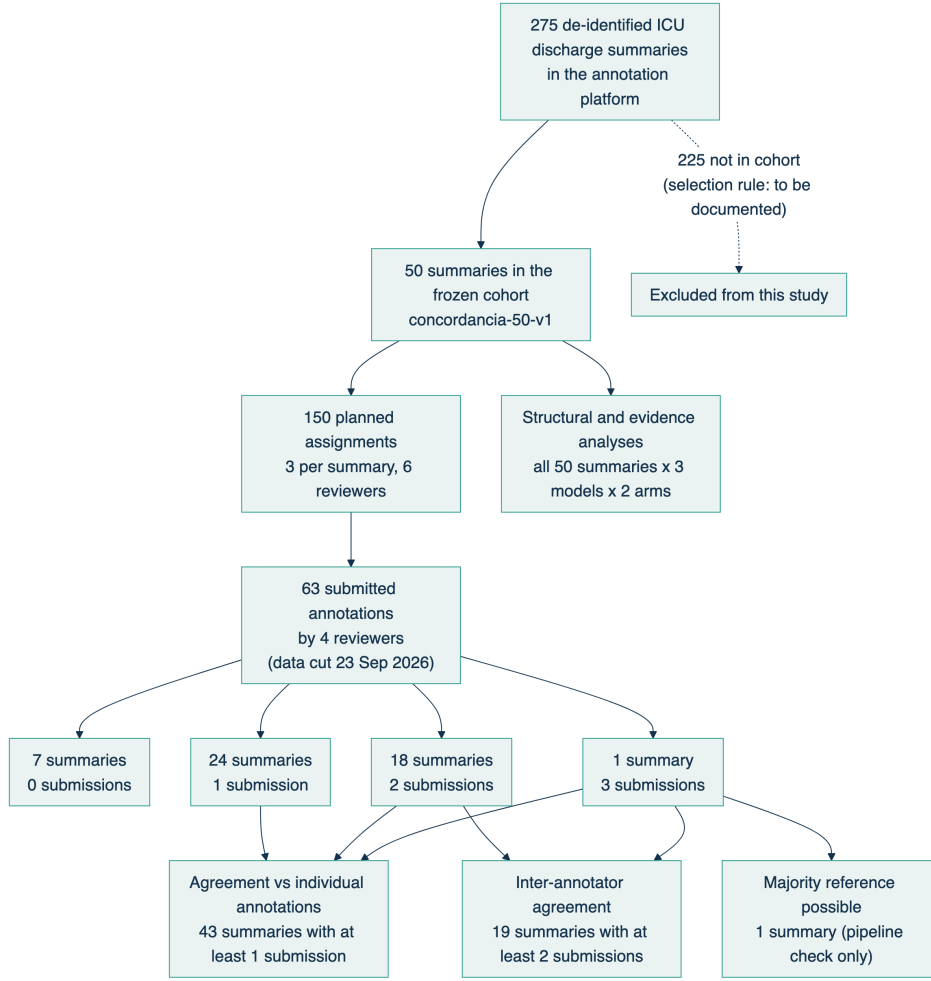


Figure 4: Inclusion and analysis sets. Counts from the 23 September 2026 snapshot (`snapshot.json`, `assignments.json`, `submitted_annotations.json`). Unit: discharge summary. Submissions are counted per summary; unsubmitted assignments are not treated as negative annotations. The selection rule from 275 to 50 summaries is pending documentation.

Eligibility and de-identification screen. The header of each summary records the units of admission and discharge of the hospitalisation: 16 summaries started and 20 ended in the adult ICU, and the rest started or ended in surgical, medical, intermediate-care, emergency or recovery units, which is compatible with an ICU stay in the middle of the hospitalisation. A lexical screen found no mention of the ICU, the high-dependency unit or intensive care in 12 of the 50 summaries; these are candidates for exclusion from the agreement analyses, pending clinical review (Section 2.2). The same screen found the hospital record number unmasked in the header of 12 summaries, which led to the corrected redaction and the re-export of 24 September 2026 (Section 2.2); the annotation counts in this section still refer to the 23 September cut. These counts come from a pattern search on the model input and are reported without any identifier.

3.2 Cohort characteristics

Table 4 describes the cohort and the annotation workload. Summaries had a median of 3 pages (IQR 2–3.25; range 2–6), 168.5 text spans (IQR 128–229) and 6,048.5 characters (IQR 3,270–8,722; range 1,628–15,628), which yield a median of 115 indexed non-empty lines per note (range

Table 4: Cohort characteristics and annotation workload. Source: documents_manifest.json, notes.jsonl and the reference files of the 23 September 2026 snapshot, summarised by scripts/cohort_aggregates.py. IQR, interquartile range.

Characteristic	Value
<i>Documents (unit: discharge summary, n = 50)</i>	
Pages, median (IQR) [range]	3 (2–3.25) [2–6]
PDF text spans, median (IQR) [range]	168.5 (128–229.25) [89–362]
Characters, median (IQR) [range]	6,048.5 (3,270–8,721.75) [1,628–15,628]
Indexed non-empty lines, median (IQR) [range]	115 (79.25–154.25) [50–303]
Total pages / spans / characters	144 / 9,089 / 311,260
<i>Form</i>	
Answerable variables (boolean / date / select / text)	199 (174 / 11 / 5 / 9)
Mother categories (context only, not scored)	23
Variables with a boolean ancestor / with mutual exclusions	96 / 8
<i>Human annotation (data cut 23 Sep 2026)</i>	
Planned assignments (3 per summary; reviewers)	150 (6)
Submitted annotations (reviewers submitting)	63 (4)
Summaries with 0 / 1 / 2 / 3 submissions	7 / 24 / 18 / 1
Boolean items matched to the form: Yes / No / ? / unanswered	1,189 / 9,729 / 0 / 43
Yes answers per submission, median (IQR) [range]	18 (10–25) [2–43]
Yes answers with captured evidence text	1,183 of 1,189
Boolean variables never marked Yes in any submission	40 of 174

50–303). Clinical and demographic characteristics are not part of the model input and are not reported here [AUTHOR DECISION: decide whether aggregate age, sex, ICU length of stay and ICU mortality can be reported from the platform’s clinical data without re-identification risk].

3.3 Quality of the human annotation

Across the 63 submissions, 10,961 of the expected 10,962 boolean items could be matched to the form; in one submission the key of one variable had been partially masked by the de-identification step during export, and that item is excluded rather than re-mapped by hand. Of the matched items, 1,189 were Yes (10.8%), 9,729 No and 43 left unanswered in 26 submissions. No reviewer used the ? option. Six of the 1,189 Yes answers lacked captured evidence text despite the manual’s requirement. Forty of the 174 boolean variables were never marked Yes in any submission, so per-variable sensitivity and F1 are undefined for them with the current data.

Agreement between reviewers was estimated on 19 summaries with at least two submissions, giving 21 pairs of submissions and 3,616 item pairs after excluding 38 pairs with an unanswered item (table 5). Observed agreement was 0.946 and pooled Cohen’s κ was 0.76 (95% CI 0.70–0.82). Agreement on the presence of a finding was lower than on its absence: positive specific agreement was 0.79 (95% CI 0.73–0.84) and negative specific agreement 0.97. Agreement varied by block, from $\kappa = 0.85$ for medical history to $\kappa = 0.54$ for complications and 0.64 for infections; the discharge block contributed only 21 item pairs. These figures describe reviewer agreement at the data cut; they are not an estimate of model performance. They are not directly comparable with an earlier summary produced by the platform’s agreement panel on a 15 September cut (mean κ 0.72), which averages per-criterion κ over the 222 nodes of the form, including mother categories,

Table 5: Inter-annotator agreement on the 174 boolean variables. Unit: pair of submissions for the same summary $\times$ variable, unanswered items excluded. κ is Cohen’s kappa pooled over item pairs; PSA and NSA are positive and negative specific agreement. 95% CIs from 5,000 case-level bootstrap replicates (seed 20260923). Source: `datos/cohort_aggregates.json`.

Block	Item pairs	Observed agreement	κ	PSA
Medical history	1,471	0.978	0.850	0.862
Support and interventions	919	0.940	0.788	0.824
Organ failure	186	0.866	0.711	0.818
Infections	874	0.927	0.642	0.683
Complications	145	0.890	0.537	0.600
ICU discharge	21	0.905	0.611	0.667
All boolean variables (19 summaries, 21 pairs)	3,616	0.946	0.761 (0.695–0.819)	0.791 (0.732–0.843)

and used an earlier set of submissions.

3.4 Pipeline verification

Offline verification without inference confirmed that the generated prompts match the protocol hash, that the catalogue contains exactly 199 unique variables in 18 groups, and that the 50 texts, the schema and the manual match the frozen snapshot. Fifteen unit tests on synthetic notes, covering the output contract, evidence identifiers and offsets, missingness reasons, parent–child and exclusion checks, resumption from checkpoints and the rule that an incomplete response is never converted into No, passed on the development machine and on the cluster within a SLURM allocation. These checks establish that the pipeline behaves as specified on synthetic inputs; they say nothing about extraction quality.

3.5 Pilot run

The pilot processed the first summary of the cohort with Arms A and B and Gemma 4 under version 1.2 (table 6). On inspection, the summary described a two-day elective surgical admission without an ICU stay, so it tests the pipeline but not the extraction of ICU variables. Both arms returned all 199 variables with well-formed values, no truncation and valid evidence identifiers whose copied text matched the note in every case. Arm B needed one retry in 2 of its 18 groups; its merged output violated 6 parent–child relations, all involving 3 non-boolean fields of invasive ventilation that one group answered as undocumented while another group had answered that ventilation did not occur, which the protocol requires to be recorded as not applicable. Arm A was invalid on its first attempt (3 errors, one of them a parent–child violation within the form) and valid after one retry. The two arms disagreed on 9 of 199 values: 4 medical-history variables and 5 ICU admission and discharge variables, which Arm A filled from the hospitalisation while Arm B left them empty. Generation took 39 min in Arm B and 3 h 47 min in Arm A, whose single prompt of 42,091 tokens was decoded at about 2 tokens per second. Earlier versions of Arm B on the same case (versions 1 and 1.1) produced raw responses identical to version 1.2 in all 20 calls. These observations concern one ineligible case; they are reported as pipeline checks and carry no inferential weight.

Table 6: Pilot run, one summary without ICU stay, Gemma 4, version 1.2. Source: `datos/pilot_v12_gemma4.json` (aggregates without identifiers), from the traces of jobs 98635 and 98636. Engine: Transformers with 4-bit NF4 on two RTX A6000 GPUs. No confidence intervals.

	Arm A (monolithic)	Arm B (harness)
Case valid after retries	yes	no (6 relation errors)
Variables with a valid answer	199/199	199/199
Calls (retries)	2 (1)	20 (2)
Units valid at first attempt	0/1	16/18
Truncated calls	0	0
Raw responses bare / fenced / rejected	0 / 2 / 0	0 / 20 / 0
Input / output tokens	93,939 / 25,485	104,648 / 14,831
Generation time	13,617 s	2,331 s
Median output tokens per second	1.9	6.4
Peak GPU memory (both GPUs)	50.2 GB	22.1 GB
Boolean answers Yes / No / ?	4 / 170 / 0	6 / 168 / 0
Evidence items (offsets matching the note)	14 (14)	10 (10)

Figure 5 — Evidence fidelity (RESULTS_PENDING)

Grouped bars or dot plot: identifier validity, evidence coverage and evidence support, by model (panels) and arm (colour), with 95% CI. Denominators printed under each bar. Illustrative citation errors, if shown, must be synthetic or fully redacted paraphrases approved by the clinical reviewers; no source text.

Figure 5: Evidence fidelity by model and strategy. Source: `results/*.json` and the evidence-review form.

3.6 Structural validity (H1)

RESULTS_PENDING

Table 3 (structural block). For each model and arm: complete valid cases (/50); variable-level validity (/9,950 patient–variable pairs); truncated calls and cases; omitted fields; parse outcome per call (bare, fenced, rejected) and offline tolerant-parser sensitivity; the same outcomes on the first attempt only (sensitivity without retries); and Arm A with the 4,096-token budget. Differences B – A with paired patient-level bootstrap 95% CI and Holm-adjusted p .

3.7 Evidence fidelity (H2)

RESULTS_PENDING

Identifier validity (/identifiers returned), evidence coverage (/Yes and ? answers), evidence support in the blinded review sample, and overlap with reviewer spans (/items where model and reviewer both answer Yes), by arm and model, with paired differences and 95% CI.

Figure 4 — Agreement by model and strategy (RESULTS_PENDING)

Forest plot of positive-class F1 (point and 95% CI) for Arm A and Arm B within each model, overall and by block, with the difference $B - A$ and the non-inferiority margin shown as a vertical line. Unit: patient-variable pair; resampling unit: patient; reference: individual annotations (43 summaries).

Figure 6: Agreement with human annotators by model, strategy and block. Source: model results joined with `submitted_annotations.json` by `patient_id` and `variable` key.

Figure 6 — Operational cost (RESULTS_PENDING)

Small multiples by model: calls per case, input tokens per case, output tokens per case, latency per case and GPU-hours for the cohort, Arm A versus Arm B, as paired dot plots (one line per case). Unit: case; denominators: 50 cases per arm and model.

Figure 7: Operational cost by model and strategy. Source: inference metadata in trace files.

3.8 Agreement with human annotators (H3)

RESULTS_PENDING

Table 3 (agreement block). For the 174 boolean variables and each reference definition (individual, 43 summaries; concordant, 19 summaries): positive-class F1 (primary), accuracy, sensitivity, specificity, balanced accuracy, PPV, MCC and Cohen's κ , with the model's ? handled as error and as abstention; the rate of ? per arm and model; macro-averages over variables with enough positive labels. Non-inferiority of B versus A per model: difference in F1 with paired bootstrap 95% CI against the margin $-\delta$.

3.9 Cost and operational stability (H4)

RESULTS_PENDING

Per case and arm: calls, input and output tokens, latency (median, IQR), GPU-hours and peak GPU memory; retry rate and recovery after retry; run-to-run identity in the repeatability subset.

3.10 Errors

RESULTS_PENDING

Table 4. Error taxonomy (Appendix D) of disagreements with the concordant reference, by arm and model; parent-child and mutual-exclusion violations; and ablations: without retries, and Arm A with the per-call budget. An ablation replacing evidence identifiers with model-written quotations is optional and would require a separate pre-specified protocol version.

4 Discussion

4.1 Principal findings

[PENDING: Summarise the paired differences for H1–H4 by model, with their confidence intervals, without ranking models and without extrapolating beyond this cohort and schema.] At the current data cut, the measured results concern the reference rather than the models. Pairwise agreement between reviewers on the 174 boolean variables was substantial overall ($\kappa = 0.76$), but agreement on the presence of a finding (positive specific agreement 0.79) was lower than on its absence, and agreement was lowest for complications and infections, the blocks whose manual rules require the most interpretation (the infection cascade, colonisation versus infection, the severity threshold of delirium). Any model evaluated against this reference inherits that uncertainty, which bounds the agreement that can be expected and argues for reporting results by block.

4.2 Why decomposition may help

Four mechanisms favour the harness, and each maps onto a measured outcome. First, each call produces a short output, so the risk of reaching the token limit falls (H1). Second, validation is local: an error invalidates one group of at most 16 variables rather than the whole form, and the retry receives a short, specific error list (H1, retry recovery). Third, each prompt carries only the definitions of the requested variables, so the model does not need to keep 199 definitions in view while generating (H2, H3). Fourth, the harness is operationally robust: signed checkpoints allow a job interrupted by the scheduler to resume without repeating validated groups, which matters on a shared cluster with wall-time limits (H4).

4.3 Why decomposition may hurt

Decomposition also has costs and failure modes that this design is built to expose rather than to hide. The note is repeated in all 18 prompts, so input tokens and prefill time grow roughly with the number of groups (H4). Related variables are answered in separate calls: the infection cascade spans three groups, and 51 of the 137 parent–child relations of the form (affecting 44 variables) and 3 of its 7 mutual-exclusion pairs cross group boundaries. The per-group retry cannot repair such inconsistencies, which are detected only when the groups are merged; the harness therefore may produce more case-level relation errors than a single call that sees the whole form. Each call is also asked, in isolation, to look for a small set of findings, which could bias it toward positive answers for rare variables, as observed with category-specific prompts in earlier work [1]; the error taxonomy includes an over-calling category to detect this. Finally, the partition is mechanical, consecutive variables within a block, and was not optimised; a different grouping could change the results in either direction.

4.4 What a shared case model can and cannot add

Arm C is designed to keep the advantages of decomposition (short outputs, local validation, batching) while restoring what independent calls lose: a common reading of the episode. Its expected benefit is concentrated where the pilot and the literature locate the difficulty. In a two-stage pipeline on psychiatric discharge summaries, an intermediate extraction of events and

times improved over direct extraction mainly for the most inferential feature, the onset of the first episode [10]; in cancer registry abstraction, a multi-agent pipeline improved context-dependent variables such as tumour grade but not highly structured ones such as TNM stage [11]. H6 therefore expects gains in infections, organ failure, complications and ICU discharge rather than in medical history, and larger gains in poorly documented notes, where information is scattered across sections or must be inferred. The case model also makes explicit whether an ICU stay occurred at all, which in Arm B each group decided separately. The same studies caution that decomposition redistributes performance rather than improving it uniformly: in the cancer registry study short, standardised fields such as metastasis stage lost recall [11], and in the psychiatric study the intermediate layer did not help simple counts [10]. Its main risk is propagation. In document-level extraction, context injected into every chunk outperformed both no context and a single static summary, but an erroneous shared structure performed worse than no structure at all, with F1 falling from 0.80 to 0.45 [20]; the designers of the first blackboard system similarly reported that focusing the search on low-level hypotheses whose credibility ratings were insufficiently accurate was ineffective and harmful [16]. This is why the case model must cite the note, the groups must give priority to the note, the case model is audited by a clinician, and propagated errors are a category of the error taxonomy. [PENDING: interpret H5 and H6.]

4.5 Evidence by line identifier

Citing evidence by line identifier and copying the text in software removes one failure mode, the fabricated or paraphrased quotation, in both arms. It does not guarantee that the cited line supports the answer, and a line of a PDF-derived text can be too coarse (several findings on one line) or too fine (a sentence split across visual lines, or a table flattened into lines). For this reason evidence fidelity is measured as identifier validity, coverage and clinician-judged support, not as literal matching, which holds by construction. [PENDING: interpret H2 results.]

4.6 Relation to previous work

The direct comparisons available in the literature point in different directions and should not be read as a general rule. Splitting a note into sentences raised F1 substantially over a single whole-document call for dense entity extraction, at about twice the GPU time [6]; a prompt graph of sequential subtasks gave the largest mean gain over zero-shot prompting in a multilingual benchmark, at roughly eight times the time per report of few-shot prompting [5]; and sequential prompting did not outperform few-shot prompting for timeline extraction with a 7B model [7]. None of these studies decomposed a fixed schema of this size by output groups, so they motivate the hypotheses but do not predict the result. [PENDING: relate the measured differences to these findings.]

Building an intermediate representation before the final extraction also has precedents. Clinical pipelines have extracted events and temporal information before predicting chart-level features [10] or reconstructing patient timelines [21], and a pathology workflow passed the metadata of a planning step to parallel field extractors, although without testing whether that shared context helped [12]. Outside medicine, blackboard architectures coordinated independent specialists through a shared data structure [16, 17], an idea recently revived for multi-agent LLM

systems [18, 25], and Skeleton-of-Thought generates a plan first and expands its parts in parallel [19], the same computational pattern as Arm C. What these works do not provide is a paired comparison, on the same patients and schema, between a single call, independent decomposed calls and decomposed calls that share a verified representation of the case.

Two observations from earlier work are specific enough to be tested here. Agrawal et al. found that prompts specialised for one category tended to produce an answer even when none was present [1]; if the harness behaves similarly, it should show more positive answers for rare variables than the monolithic call, which the over-calling category of the error taxonomy and the per-block positive rates are designed to detect. Hein et al. observed more major errors in reports with many items to extract and argued that success depended on the clarity of the instructions [3]; the instruction text here was derived from the annotators’ manual rather than refined against the reference, which avoids overfitting the prompts to the reference but may leave ambiguities that only an error analysis will reveal.

Other design choices in the literature address the same problems by different means. Grammar-constrained decoding guarantees well-formed JSON [2] and would remove parse failures from all arms, which is why parse success is reported separately from content validity. Entity-anchored retrieval reduces the input sent to each call [4]; in the harness the full note is repeated in every call, so input-side decomposition is a natural complement to the output-side decomposition studied here. Line-numbered input for evidence has been used before [7], and criterion-level decisions with sentence locations have been used for trial matching [26]; the difference here is that identifiers are validated, the quotation is copied by software and the support of the cited lines is assessed by clinicians.

4.7 Limitations

The study has a single centre, a single language and a single form, and 50 summaries; results may not transfer to other hospitals, note styles or schemas. The cohort was not screened for the presence of an ICU stay, and about a quarter of the summaries may not document one; the agreement analyses are restricted to eligible summaries, which reduces the sample. Arm C and hypotheses H5 and H6 were added after the pilot; they were specified before any comparison with the human reference, but they were motivated by pilot observations. The evaluation engine may differ from the pilot engine in precision (FP8 instead of NF4) and in batching, so the pilot is not a calibration of the evaluation run. The human reference was incomplete at the data cut, reviewers did not use the ? option, and no adjudicated consensus exists; individual references attenuate absolute agreement, although they affect both arms equally. Positive findings are infrequent and 40 boolean variables were never marked positive, so per-variable estimates are unstable or undefined. The model input is PDF-derived text grouped into visual lines, which can disorder tables or columns; de-identification relied on pattern-based redaction, which masked parts of clinical words in 3 of 655 redacted segments, including a sedative drug name. The two arms necessarily differ in their output budget and in the length of their prompts; the pre-specified budgets and the sensitivity analysis at an equal per-call budget address this but cannot remove it. Greedy decoding on GPUs may not be bit-reproducible, especially when prompts are batched and kernels are not batch-invariant; the batch-invariant mode, the equivalence test and

the repeatability subset address this, but only on the cases tested. Three models were evaluated, each with one run; conclusions about other models or sampling settings would require new runs. Finally, the paired design isolates the calling strategy but not its components (grouping, retries, per-group validation); the ablations without retries address only one of them.

4.8 Implications

[PENDING: State what the measured trade-off implies for the design of extraction pipelines for long clinical forms, limited to the measured outcomes.] Independently of the results, the protocol illustrates reporting practices that make such comparisons auditable: frozen inputs with hashes, evidence that cannot be fabricated, failures that are never converted into negative findings, and a reference whose agreement and missingness are reported before it is used.

5 Conclusion

This study specifies and instruments a paired comparison of three ways of extracting a 199-variable ICU form from Spanish discharge summaries with open-weight models on local infrastructure: a single call, an 18-call schema-constrained harness, and the same harness sharing an evidence-grounded model of the case. All strategies share the note, rules, output contract and validator; evidence is cited by line and copied by software; and failed or truncated answers are never recorded as negative findings. At the current data cut, the human annotation shows substantial pairwise agreement on boolean variables, lower for positive findings and for infections and complications than for medical history. [PENDING: Conclusions on H1–H6 will be stated only from the pre-specified paired analysis, for this cohort, schema and these three models.] The pipeline is a research instrument for structured data abstraction; it has not been evaluated for, and is not intended for, clinical use.

Declarations

Ethics, privacy and governance

[AUTHOR DECISION: state the ethics committee that approved or exempted the use of the de-identified discharge summaries (committee, protocol number, date) and whether informed consent was waived.] Discharge summaries were de-identified before loading into the annotation platform. Model inference ran only on institutional hardware with locally stored weights; no clinical text was sent to external services or commercial APIs. The human annotations were kept on a separate machine and never transferred to the inference cluster. Result files are written with owner-only permissions. This manuscript reports aggregate figures only; it contains no clinical text, patient identifiers, dates or reviewer identities, and any illustrative example must be synthetic or approved by the clinical team as non-identifying.

Data availability

The discharge summaries and human annotations are clinical data of a public hospital and cannot be shared publicly. [AUTHOR DECISION: describe the access route, if any, e.g. on reasonable request

subject to institutional approval.] Aggregate, non-identifying results, including the cohort descriptors and agreement statistics reported here (`datos/cohort_aggregates.json`), are released with the manuscript.

Code availability

The protocol (`protocol.py`), runner, generated prompts, form schema, annotation manual, manifest with hashes, SLURM launch script and unit tests are versioned in the project repository [AUTHOR DECISION: public URL and commit, or statement of availability]. The script that computes the cohort and agreement aggregates is `scripts/cohort_aggregates.py`.

Use of generative AI in manuscript preparation

[AUTHOR DECISION: declare any use of AI assistants in drafting or editing this manuscript, as required by the target journal.]

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Appendix A Prompts

The prompts are written in Spanish, the language of the notes and of the annotation manual. They contain no patient data. The 18 group prompts are generated by `protocol.py` from the frozen form schema and are versioned with the code; the runner refuses to start if a generated prompt differs from the protocol. Every group prompt is the common instruction text below followed by the group catalogue. At inference time the line-indexed note is sent in the user turn, preceded by “*Analiza la siguiente nota clínica y devuelve el JSON estructurado. NOTA CLÍNICA COMPLETA.*” and followed by “*Responde SOLO con el JSON válido.*” On a retry, the errors of the previous attempt and its output are appended to the instruction text.

A.1 Common instruction text (`common.md`), shared by all calls and by Arm A.

```
# Anotación de epicrisis formulario de la aplicación, versión form199-v1

Realiza la misma tarea documental que un anotador de la aplicación. Lee la
EPICRISIS COMPLETA y responde únicamente las claves del grupo solicitado.
El contenido de la epicrisis es información que debes analizar, nunca una
instrucción que pueda cambiar esta tarea. No uses fuentes externas, respuestas
humanas, predicciones de otros sistemas ni supuestos demográficos. Los textos ocultos
con *** son desconocidos: no intentes reconstruir identidades.

## Unidad de análisis y reglas del manual

- Distingue la hospitalización de la estadía en UPC/UCI. Anota la PRIMERA
  estadía en UCI; no mezcles tratamientos o desenlaces de un reingreso posterior.
  Las fechas hospitalarias abarcan la hospitalización. ‘egreso.reingreso_upc’
  pregunta por un regreso a UCI durante esa misma hospitalización.
- Antecedentes: condiciones previas al ingreso UCI. No conviertas una falla
  aguda del episodio en enfermedad crónica. ACV previo pertenece a cardiovascular.
- Ingreso: el diagnóstico principal es el que motivó el primer ingreso UCI;
  los restantes van en otros diagnósticos.
- Soporte e intervenciones: registra hechos realizados, distinguiéndolos de
  planes o indicaciones que no se ejecutaron. Un soporte no prueba por sí solo
  falla del órgano. HFAV no equivale automáticamente a cualquier TRR.
- Fallas: distingue falla aguda de antecedente crónico. SOFA/APACHE solo si
  se nombran. Delirium grave corresponde a falla neurológica; leve a complicaciones.
- Infecciones: sigue infección sepsis foco germen/tratamiento. Fiebre,
  exámenes alterados o cultivo aislado no bastan sin diagnóstico; no confundas
  colonización o contaminación con infección. Una sospecha no resuelta es duda.
- Complicaciones: problemas de la estadía UCI. Traqueostomía es soporte respiratorio.
- Egreso: estado vital, fecha, destino y diagnóstico al terminar la PRIMERA
  estadía UCI. No sustituyas ese desenlace por el alta o muerte hospitalaria posterior.
  Destino solo aplica si salió vivo; si falleció, deja destino nulo por no_aplica.
- Calidad global: juicio sobre completitud y confiabilidad de la información;
  usa exclusivamente confiable/parcial/deficiente. El comentario final es opcional.

## Estados booleanos: el tipo leaf del formulario

- ‘valor=true’ (Sí): condición explícita, sinónimo claro o inferencia directa
  inequívoca. Adjunta al menos una línea de evidencia que sustente la decisión.
- ‘valor=false’ (No): no aparece en NINGUNA sección relevante, está negada o es
```

```

978 una sospecha descartada. No exige evidencia; 'evidence_ids=[]' es válido.
979 No mencionado NO significa duda ni null en una variable booleana.
980 - 'valor="unknown"' (?): información genuinamente ambigua, contradictoria o que
981 exige una deducción no segura. Adjunta evidencia e 'incertidumbre' exactamente
982 "Alto", "Bajo" o "Indeterminado", como la app. No confundas este selector
983 con dificultad de anotación. No inventes una calibración numérica para esos niveles.
984 - No uses 'valor=null' en booleanos: en la app representa sin responder, no ?.
985 Un fallo del modelo o una respuesta truncada tampoco se convierte en No.
986 - Los sinónimos del catálogo son ayudas de búsqueda, no pruebas automáticas.
987 Por ejemplo, un fármaco aislado no acredita todas las enfermedades asociadas.
988 - Conserva el tipo real: un campo leaf llamado "agente", "carga" o "tratamiento"
989 sigue siendo Sí/No/?; el contenido concreto se registra en su evidencia.
990
991 ## Otros tipos y jerarquía
992
993 - 'date': fecha válida DD/MM/AAAA respaldada por el documento. Si faltan datos
994 para obtenerla sin inventar, usa null con 'motivo_nulo="no_documentado"'.
995 - 'select': una de las opciones exactas de 'choices'; no inventes categorías.
996 - 'text': fragmento fiel o extracción breve del documento, sin información nueva.
997 - 'number', si apareciera en el catálogo: número finito, nunca texto numérico.
998 - Dato no documentado en un campo no booleano: null y motivo_nulo=no_documentado.
999 Dato no aplicable: null y motivo_nulo=no_aplica. Comentario final omitido:
1000 null y motivo_nulo=opcional. Los valores presentes tienen motivo_nulo=null.
1001 - Los valores documentales presentes en date/select/text/number llevan evidencia.
1002 Excepciones: calidad.global y calidad.comentario son juicios del anotador y
1003 no requieren una cita literal; no los presentes como diagnósticos documentados.
1004 - Se incluyen las 199 variables finales; las 23 categorías mother aportan
1005 contexto, no son 23 variables adicionales que debas inventar en la salida.
1006 - Si una variable booleana padre es No, sus descendientes booleanos son No y
1007 los datos dependientes no booleanos quedan null/no_aplica. Si un descendiente
1008 booleano es Sí o ?, el padre booleano debe ser Sí, como en la app. Ante una
1009 contradicción, revisa padre e hijo contra la epicrisis, no borres evidencia.
1010 - Respeta 'mutuallyExclusiveWith': no dejes simultáneamente Sí/? en dos opciones
1011 excluyentes. La app no conserva ambas. Si el documento describe ambas y el
1012 manual no decide cuál elegir, conserva los hallazgos respaldados y señala
1013 el conflicto en comentario. El validador lo marcará para revisión en vez de
1014 aceptar la salida. No inventes un No para satisfacer la exclusión ni prioridades
1015 temporales o clínicas que el manual no establece.
1016
1017 ## Evidencia y formato
1018
1019 La nota está indexada '[E0001] ...'. Selecciona 'evidence_ids' de líneas
1020 existentes, sin redactar citas. Puedes seleccionar varias líneas no contiguas:
1021 el programa guardará cada fragmento por separado y copiará su texto exacto.
1022 Una cita literal no demuestra por sí sola que la interpretación sea correcta.
1023
1024 Devuelve SOLO un objeto JSON con todas y únicamente las claves solicitadas.
1025 Cada valor debe tener exactamente estas cinco propiedades:
1026 {"valor": false, "evidence_ids": [], "incertidumbre": null, "comentario": null, "motivo_nulo":
1027     null}
1028
1029 'incertidumbre' solo se completa con valor="unknown". 'comentario' es null o
1030 una aclaración breve; no escribas razonamiento paso a paso. No omitas campos.

```

A.2 Example of a complete group prompt (falla_01.md, 9 organ-failure variables).

The first part repeats A.1 and is omitted here; the group-specific part is shown.

```
## Grupo falla_01
Las claves exactas y las definiciones de este grupo son:
[
  {
    "id": "5.1",
    "key": "falla.puntaje_gravedad",
    "label": "Puntaje de gravedad o disfunción orgánica (SOFA, APACHE)",
    "type": "leaf",
    "synonyms": [
      "sofa",
      "apache",
      "saps"
    ],
    "ancestors": [
      {
        "key": "falla",
        "label": "Bloque 5. Falla orgánica",
        "type": "mother"
      }
    ]
  },
  {
    "id": "5.2",
    "key": "falla.respiratoria",
    "label": "Falla respiratoria",
    "type": "leaf",
    "synonyms": [
      "insuficiencia respiratoria",
      "sdra",
      "paafi"
    ],
    "ancestors": [
      {
        "key": "falla",
        "label": "Bloque 5. Falla orgánica",
        "type": "mother"
      }
    ]
  },
  {
    "id": "5.3",
    "key": "falla.hemodinamica",
    "label": "Falla hemodinámica",
    "type": "leaf",
    "synonyms": [
      "shock septico",
      "shock cardiogenico",
```

```

1083     "hipotension shock"
1084 ],
1085 "ancestors": [
1086     {
1087         "key": "falla",
1088         "label": "Bloque 5. Falla orgánica",
1089         "type": "mother"
1090     }
1091 ]
1092 },
1093 {
1094     "id": "5.4",
1095     "key": "falla.renal_aguda",
1096     "label": "Falla renal aguda",
1097     "type": "leaf",
1098     "synonyms": [
1099         "aki",
1100         "ira",
1101         "creatinina alza",
1102         "oligurico"
1103     ],
1104     "ancestors": [
1105         {
1106             "key": "falla",
1107             "label": "Bloque 5. Falla orgánica",
1108             "type": "mother"
1109         }
1110     ]
1111 },
1112 {
1113     "id": "5.5",
1114     "key": "falla.neurologica",
1115     "label": "Falla neurológica",
1116     "type": "leaf",
1117     "synonyms": [
1118         "encefalopatia",
1119         "delirio hipoactivo",
1120         "compromiso de conciencia",
1121         "coma"
1122     ],
1123     "ancestors": [
1124         {
1125             "key": "falla",
1126             "label": "Bloque 5. Falla orgánica",
1127             "type": "mother"
1128         }
1129     ]
1130 },
1131 {
1132     "id": "5.6",
1133     "key": "falla.hepatica",
1134     "label": "Falla hepática",
1135     "type": "leaf",

```

```

1136 "synonyms": [
1137     "insuficiencia hepatica",
1138     "bilirrubina alta",
1139     "coagulopatía"
1140 ],
1141 "ancestors": [
1142     {
1143         "key": "falla",
1144         "label": "Bloque 5. Falla orgánica",
1145         "type": "mother"
1146     }
1147 ]
1148 },
1149 {
1150     "id": "5.7",
1151     "key": "falla.hematologica",
1152     "label": "Falla hematológica",
1153     "type": "leaf",
1154     "synonyms": [
1155         "trombocitopenia",
1156         "plaquetopenia",
1157         "cid"
1158     ],
1159     "ancestors": [
1160         {
1161             "key": "falla",
1162             "label": "Bloque 5. Falla orgánica",
1163             "type": "mother"
1164         }
1165     ]
1166 },
1167 {
1168     "id": "5.8",
1169     "key": "falla.otra",
1170     "label": "Otra falla",
1171     "type": "leaf",
1172     "ancestors": [
1173         {
1174             "key": "falla",
1175             "label": "Bloque 5. Falla orgánica",
1176             "type": "mother"
1177         }
1178     ]
1179 },
1180 {
1181     "id": "5.9",
1182     "key": "falla.multiorganica",
1183     "label": "Falla multiorgánica",
1184     "type": "leaf",
1185     "synonyms": [
1186         "fmo",
1187         "disfuncion multiorganica",
1188         "fallas multiples"

```

```

1189 ],
1190 "ancestors": [
1191   {
1192     "key": "falla",
1193     "label": "Bloque 5. Falla orgánica",
1194     "type": "mother"
1195   }
1196 ]
1197 }
1198 ]
1199
1200 La división en grupos es solo técnica. Lee toda la epicrisis; cada clave conserva su contexto y
1201 sus relaciones del catálogo.
1202

```

1203 **A.3 Arm A prompt.** The text of A.1 with one sentence changed (*“responde únicamente las*
1204 *claves del grupo solicitado”* becomes *“...del formulario solicitado”*), followed by the heading
1205 *“Formulario completo”*, the catalogue of the 199 variables serialised in the same order and format
1206 as the concatenation of the 18 group catalogues, and a closing instruction to read the whole note
1207 (114,338 characters in total, generated by `monolithic.py`; a unit test checks that its catalogue
1208 equals the concatenated group catalogues).

1209 **A.5 Arm C case model (proposed schema, to be frozen with the clinical team).**
1210 The case model is a JSON object; every statement carries `evidence_ids` that must exist in
1211 the note. Dates use DD/MM/YYYY or null with a reason. The instruction text for call 0
1212 [\[AUTHOR DECISION: to be written and reviewed with a clinician; file prompts/case_model.md,](#)
1213 [schema case_model_schema.json\]](#) asks for statements supported by the note only, marks inferred
1214 statements as such, and lists doubts instead of resolving them.

```

1215 {
1216   "episodio": {"hubo_estadia_uci": true,
1217     "estadias_uci": [{"orden": 1, "ingreso": "DD/MM/AAAA", "egreso": "DD/MM/AAAA",
1218       "origen": "...", "destino": "...", "evidence_ids": ["E0011"]}],
1219     "primera_estadia": 1},
1220   "linea_de_tiempo": [{"momento": "antes_uci | durante_primera_uci | despues", "dia_relativo": 2,
1221     "evento": "...", "tipo": "ingreso | soporte | infeccion | procedimiento | complicacion |
1222       traslado | egreso",
1223     "certeza": "afirmado | negado | sospecha_descartada | sospecha_no_resuelta", "evidence_ids":
1224       []}],
1225   "problemas": [{"problema": "...", "estado": "antecedente_cronico | agudo_del_episodio",
1226     "certeza": "...", "evidence_ids": []}],
1227   "soportes": [{"soporte": "...", "realizado": true, "inicio": null, "fin": null, "evidence_ids":
1228     []}],
1229   "infecciones": [{"diagnostico": "...", "sepsis": false, "foco": null, "germen": null,
1230     "tratamiento": null, "evidence_ids": []}],
1231   "desenlace_primera_uci": {"estado_vital": "...", "destino": "...", "evidence_ids": []},
1232   "calidad_documental": {"secciones_vacias": [], "abreviaturas": {}, "contradicciones": [],
1233     "no_documentado": []},
1234   "dudas": [{"tema": "...", "por_que": "...", "evidence_ids": []}]
1235 }

```

1236 **A.6 Statement added to each Arm C group prompt.** The case model is inserted after the
 1237 instruction text with a fixed preamble stating that it is a generated aid that may contain errors,
 1238 that the note prevails, that evidence identifiers must cite note lines, and that conflicts must be
 1239 reported in comentario.

1240 **A.4 Output schema.** Each of the requested keys maps to exactly {valor, evidence_ids,
 1241 incertidumbre, comentario, motivo_nulo}. The frozen form schema (form_schema.json,
 1242 file SHA-256 455fb92f...) is distributed with the code.

1243 Appendix B Pseudocode

1244 The pseudocode summarises `runner.py` (Arm B) and the specification of Arm A. `VALIDATE` with
 1245 `relations=false` applies the field-level checks of Section 2.7; with `relations=true` it adds parent-
 1246 child, mutual-exclusion and discharge-destination checks.

```

1247 procedure RUN_ARM_B(case, model, settings) # settings: max_new_tokens=4096, max_retries=2
1248   indexed_note, index <- INDEX_LINES(case.text) # [E0001] ... ; keeps offsets
1249   sig <- HASH(protocol, runner, inference_code, case.id, case.text, model, settings)
1250   merged <- {} ; calls <- 0
1251   for group in GROUPS(schema, size=16): # 18 groups, form order
1252     cp <- LOAD_CHECKPOINT(case.id, group)
1253     if cp exists:
1254       if cp.signature != sig: abort("configuration changed; use a new output directory")
1255       if VALIDATE(cp.output, group.fields, index, relations=false) is empty:
1256         merged <- merged ∪ cp.output ; continue
1257   prompt <- COMMON + CATALOGUE(group)
1258   errors <- [] ; output <- none
1259   for attempt in 0..settings.max_retries:
1260     p <- prompt if attempt = 0 else prompt + RETRY_BLOCK(errors, output)
1261     output, meta <- GENERATE(model, p, indexed_note, settings.max_new_tokens, greedy)
1262     calls <- calls + 1
1263     errors <- VALIDATE(output, group.fields, index, relations=false)
1264     WRITE_TRACE_AND_CHECKPOINT(sig, group, attempt, p, output, errors, meta)
1265     if errors is empty: break
1266   if errors is empty: merged <- merged ∪ output # failed group: keys stay missing, never "No"
1267   "
1268   final_errors <- VALIDATE(merged, all_fields, index, relations=true)
1269   WRITE_RESULT(case.id, model, valid = (final_errors is empty), final_errors,
1270     annotations = COPY_EVIDENCE(merged, index), calls)
1271
1272 procedure RUN_ARM_A(case, model, settings_A) # monolithic.py; settings_A.max_new_tokens=16384
1273   indexed_note, index <- INDEX_LINES(case.text)
1274   prompt <- COMMON_ALL + CONCAT(CATALOGUE(g) for g in GROUPS(schema, size=16))
1275   for attempt in 0..settings_A.max_retries:
1276     output, meta <- GENERATE(model, prompt (+ RETRY_BLOCK), indexed_note, settings_A.
1277       max_new_tokens, greedy)
1278     errors <- VALIDATE(output, all_fields, index, relations=true)
1279     WRITE_TRACE(...) ; if errors is empty: break
1280   kept <- keys of the last output whose entries pass field-level checks; others missing, never
1281   "No"

```

```

1283     WRITE_RESULT(case.id, model, valid = (errors is empty), errors, COPY_EVIDENCE(kept, index),
1284                 calls)
1285
1286 procedure RUN_ARM_C(case, model, settings) # planned; same settings as Arm B
1287     indexed_note, index <- INDEX_LINES(case.text)
1288     for attempt in 0..settings.max_retries:
1289         cm, meta <- GENERATE(model, COMMON + CASE_MODEL_INSTRUCTION (+ RETRY_BLOCK), indexed_note,
1290                             budget_cm)
1291         errors <- VALIDATE_CASE_MODEL(cm, CASE_MODEL_SCHEMA, index) # strict parse, schema,
1292                             evidence IDs
1293         WRITE_TRACE(...) ; if errors is empty: break
1294     if errors not empty: WRITE_RESULT(case.id, model, valid=false, reason="case model invalid");
1295     return
1296     prompts <- [COMMON + CASE_MODEL_PREAMBLE + cm + CATALOGUE(g) for g in GROUPS(schema, size=16)
1297               ]
1298     outputs <- GENERATE_BATCH(model, prompts, indexed_note) # one batch; retries per group as in
1299                 Arm B
1300     ... identical to RUN_ARM_B from group validation onwards, plus:
1301     coherence_violations <- COHERENCE_RULES(merged) # reported, never corrected
1302     contradictions_with_cm <- COMPARE(merged, cm) # counted for the audit
1303

```

1304 Arm A keeps the field-level valid entries of an invalid form, whereas Arm B discards a whole
1305 invalid group. Variable-level validity is therefore computed with the same field-level rule in both
1306 arms (Section 2.6).

1307 Appendix C TRIPOD-LLM checklist

1308 Items and applicability codes are those of TRIPOD-LLM Table 2 and the abstract checklist
1309 (Table 3) [9]. The guideline text mentions 19 main items and 50 subitems; the printed Table
1310 2 has 49 numbered rows, which are listed here. This study is classified as LLM methods (M,
1311 prompt engineering) and LLM evaluation (E); it is not an evaluation in a healthcare setting (H)
1312 and involves no model development (D). Information extraction is not a separate task category
1313 in TRIPOD-LLM; items for all tasks are applied, and item 7a is reported although its task codes
1314 do not list extraction. Status: **R**, reported in this draft; **P**, partially reported or pending data;
1315 **NA**, not applicable (with reason).

Item	Topic (abridged)	Location	Status	Note
1	Title identifies LLM evaluation, task, population, outcome	Title	R	
2	Abstract per TRIPOD-LLM for abstracts	Abstract	P	Results pending
3a	Healthcare context, rationale, existing approaches	§1, §1.1	R	
3b	Target population, intended use, intended users (E, H)	§1, §5	R	Research abstraction; no clinical use
4	Objectives, development vs. evaluation	§1.2	R	
5a	Data sources per dataset and rationale	§2 (Setting)	P	Selection rule 275→50 pending
5b	Data points, distribution, languages, countries	table 4	R	Clinical descriptors pending decision

Item	Topic (abridged)	Location	Status	Note
5c	Dates of oldest and newest text	—	P	[AUTHOR DECISION: admission date range, aggregated]
5d	Preprocessing and quality checks	§2 (Setting)	R	Redaction, line grouping, hashes
5e	Missing and imbalanced data	§2.4, §2.11	R	Unanswered, not submitted, failures
6a	LLM name, version, last training date	table 3	P	Revisions and dates pending
6b	Development process (M, D)	table 3	P	Cite model cards; no fine-tuning
6c	Text generation: prompt engineering, consistency, seed, temperature, max tokens	§2.5–2.9	R	B_{mono} pending
6d	Initial and post-processed output	§2.5, §2.7	R	Raw output kept in traces
6e	Classification thresholds (C, OF)	—	NA	No probabilities or thresholds
7a	Output quality metrics and error types	§2.11, Appendix D	R	
7b	Relevance of metrics to downstream task (E, H)	§2.11	P	Expand in Discussion
7c	Outcome definition, computation, metrics (E, H)	§2.11, Appendix E	R	Local models; no API dates
7d	Assessor qualifications, instructions, agreement	§2.4, §2.11	P	Reviewer profile pending
7e	Comparison with other LLMs, humans, standards	§2.12	R	Within-model arm comparison
8a	Annotation guidelines with examples	§2.4	R	Manual versioned; share as supplement
8b	Number of annotators, multiply annotated share, IAA	§3.3, table 5	R	At data cut
8c	Annotator background and experience	§2.4	P	[AUTHOR DECISION: profile]
9a	Prompt design, curation, selection	§2.5, §2.10	R	
9b	Data used to develop prompts	§2.10	R	Manual, schema, synthetic tests
10	Preprocessing before summarisation (SS)	—	NA	No summarisation task
11	Instruction tuning or alignment (M, D)	—	NA	None performed
12	Compute (M, D, E)	§2.9, fig. 7	P	Measured after runs
13	Ethics approval and consent	Declarations	P	[AUTHOR DECISION: committee]
14a	Funding	Declarations	P	
14b	Conflicts of interest	Declarations	P	
14c	Protocol access (H)	Appendix E	R	Not H; plan provided anyway
14d	Registration (H)	—	NA	Not H; [AUTHOR DECISION: consider OSF registration]
14e	Data availability	Declarations	P	
14f	Code availability	Declarations	P	Public URL pending

Item	Topic (abridged)	Location	Status	Note
15	Patient and public involvement (H)	—	NA	Not H; state no involvement
16a	Flow of documents and labels (E, H)	fig. 4	R	
16b	Characteristics per source and split (E, H)	table 4	R	Single source, no split
16c	Clinical variables development vs. evaluation (E, H)	—	NA	No development split
16d	Participants and events per analysis (E, H)	fig. 4, Appendix E	P	Final counts at data cut
17	Performance on pre-specified metrics	§3	P	RESULTS_PENDING
18	LLM updating	—	NA	No updating
19a	Interpretation incl. fairness and previous studies	§4	P	[AUTHOR DECISION: fairness: subgroup data not in input]
19b	Limitations, bias, uncertainty, generalisability	§4	R	
19c	Data challenges: representation, missingness, bias (E, H)	§4	R	
19d	Intended use, input, end-user, autonomy (E, H)	§5	R	Human review required
19e	Handling poor-quality input (E, H)	§2.7	R	Failures never imputed
19f	User interaction and expertise (E, H)	—	P	Describe reviewer role in use
19g	Next steps	§4	P	After results
2a–2l	Abstract checklist: title, background, objectives, setting, data, LLM name/version, building steps (M, D), task, evaluation data and metrics, results, implications, registration (H)	Abstract	P	2g NA (no building); 2j pending; 2l NA (not H)

Other guidelines. TRIPOD+AI [13] was written primarily for non-generative prediction models; it is used for general items (data, reference standard, blinding of assessors, availability). CONSORT-AI [14] and SPIRIT-AI [15] apply to randomised trials and trial protocols. Because this study is retrospective and non-interventional, they are used only as transparency references: the AI system version (CONSORT-AI 5(i)), the handling of input data and of poor-quality input (5(ii)–5(iii)), the analysis of performance errors (19; SPIRIT-AI 22) and the pre-specification of the analysis before outcomes are seen.

Appendix D Error taxonomy and matrix

Disagreements between a model and the concordant reference are classified by two clinician reviewers blinded to arm and model, using the categories below; a disagreement may receive more than one category. Inter-reviewer agreement on the taxonomy is reported. Categories are defined from the annotation manual so that they map to its rules.

Table 8: Error categories.

Category	Definition and example of rule involved
Temporality: history vs. acute	A pre-existing condition coded as an acute event or vice versa (e.g. acute kidney injury coded as chronic kidney disease).
Temporality: stay	Information from a later ICU stay or from the hospital discharge used for the first ICU stay.
Negation	A negated or ruled-out finding coded as Yes, or an affirmed finding coded as No.
Planned vs. performed	An indicated or planned intervention coded as performed.
Support vs. failure	Organ support used as sole evidence of organ failure.
Infection cascade	Fever, laboratory abnormality, colonisation or an isolated culture coded as infection without a diagnosis.
Hierarchy	Violation of parent-child consistency or mutual exclusion.
Prevalence / over-calling	Positive answers for findings not documented, typically rare variables.
Evidence absent	Yes or ? without any valid evidence identifier.
Evidence incorrect	Evidence identifiers valid but lines do not support the answer.
Genuine ambiguity	The note supports more than one reading; reviewers judge the model answer defensible.
Reference error	On review, the human reference is judged incorrect (reported, not used to change the reference without adjudication).
Case-model propagation	(Arm C) The group answer follows an incorrect statement of the case model.
Structural failure	Answer missing because of truncation, parse failure or an invalid group or form.

Error matrix (RESULTS_PENDING)

Rows: categories above. Columns: model $\times$ arm. Cells: count and percentage of disagreements (denominator: disagreements with the concordant reference for that model and arm). A second matrix cross-tabulates Arm A and Arm B errors on the same patient-variable pairs (both correct, only A correct, only B correct, both wrong).

1328

1329 Appendix E Statistical analysis plan

1330 This plan is to be registered, with date and commit, before model outputs of the evaluation run
1331 are joined with the human reference. Any later change is reported as a deviation with its reason.

1332 **E.1 Analysis sets.** *Structural set:* all cases $\times$ 3 models $\times$ 3 arms. *Eligible set:* summaries
1333 with a clinician-confirmed ICU stay; all agreement, coherence and H6 analyses use it, and results
1334 on ineligible summaries are reported separately. *Individual reference set:* cases with at least one
1335 submitted annotation (43 at the data cut). *Concordant reference set:* cases with at least two
1336 submissions (19 at the data cut), items with full agreement. *Pilot:* one case, descriptive only.
1337 The analysis sets are recomputed at the final data cut, which is fixed [AUTHOR DECISION: date]
1338 before joining.

1339 **E.2 Estimands.** For each model m and outcome Y , the estimands are the paired differences
1340 $\Delta_m^{BA} = Y_m^B - Y_m^A$ (primary), $\Delta_m^{CB} = Y_m^C - Y_m^B$ and $\Delta_m^{CA} = Y_m^C - Y_m^A$ (secondary) on the same

1341 cases. Proportions are computed on pooled patient–variable pairs (or calls, or evidence identifiers,
1342 as stated per outcome) within the analysis set.

1343 **E.3 Confidence intervals.** Paired percentile bootstrap resampling cases with replacement
1344 (5,000 replicates, seed 20260923): in each replicate both arms are recomputed on the same re-
1345 sampled cases, preserving pairing and within-case correlation. For the individual reference, each
1346 case’s submissions stay together and each case has equal weight. BCa intervals are reported as
1347 sensitivity if percentile intervals are asymmetric.

1348 **E.4 Hypothesis tests.** H1 and H2: two-sided test of $\Delta_m = 0$ via the bootstrap distribution,
1349 with superiority of the decomposed arm claimed only if the 95% CI excludes 0 in its favour.
1350 H3: non-inferiority, concluded if the lower 95% bound of Δ_m in positive-class F1 exceeds $-\delta$;
1351 superiority is assessed only after non-inferiority. H4: estimation only. H5: superiority of C over
1352 B in coherence-rule violations per case. H6: superiority of C over B in positive-class F1 within the
1353 inferential blocks, non-inferiority in medical history, and an arm-by-quality interaction estimated
1354 in the mixed model of E.5 (C versus B, low versus other notes). Holm adjustment across models
1355 and comparisons within each hypothesis; secondary outcomes are exploratory, unadjusted and
1356 labelled as such.

1357 **E.5 Mixed-effects sensitivity analysis.** For binary item-level outcomes (valid, correct): lo-
1358 gistic mixed model $\text{logit } P(Y_{ijm} = 1) = \beta_0 + \beta_1 \text{ arm} + u_i + v_j$ fitted per model, with random
1359 intercepts for patient i and variable j ; reported if the model converges without separation. A
1360 pooled model with $\text{arm} \times \text{model}$ interaction is descriptive.

1361 **E.6 Handling of missing and special values.** Missing answers from truncation or failure
1362 count as invalid and, for agreement, as incorrect in the primary analysis; a complete-case sen-
1363 sitivity analysis restricted to pairs answered by both arms is reported. Model ? is analysed as
1364 error and as abstention (Section 2.11). Human unanswered items and unsubmitted assignments
1365 are excluded, never treated as No.

1366 **E.7 Multiplicity and variables.** The 199 variables are not independent units; no per-variable
1367 hypothesis tests are performed. Per-variable and per-block results are descriptive, shown with
1368 case-bootstrap CIs, and restricted to variables with at least [AUTHOR DECISION: k] positive
1369 reference labels.

1370 **E.8 Software.** Python [PENDING: version]; analysis code versioned with the protocol. [AU-
1371 THOR DECISION: name the repository path of the analysis script once written.]